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Review

Survey of Architectural Floor Plan Retrieval Technology Based on 3ST Features

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Abstract: Feature retrieval technology for building floor plans has garnered significant attention in recent years due to its critical role in the efficient management and execution of construction projects. This paper presents a comprehensive exploration of four primary features essential for the retrieval of building floor plans: semantic features, spatial features, shape features, and texture features (collectively referred to as 3ST features). The extraction algorithms and underlying principles associated with these features are thoroughly analyzed, with a focus on advanced methods such as wavelet transforms and Fourier shape descriptors. Furthermore, the performance of various retrieval algorithms is evaluated through rigorous experimental analysis, offering valuable insights into optimizing the retrieval of building floor plans. Finally, this study outlines prospective directions for the advancement of feature retrieval technology in the context of floor plans.

Keywords: floor plan feature retrieval; 3ST features; wavelet transforms; Fourier shape descriptions

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1. Introduction

In the field of architectural design, architectural blueprints and floor plans are two fundamental concepts; however, their meanings and functions are frequently conflated. Architectural blueprints encompass the comprehensive design documentation of a construction project, typically comprising floor plans, elevations, sections, and other detailed drawings that illustrate the building's structural, electrical, and plumbing systems. In contrast, floor plans constitute a specific component of architectural blueprints, primarily depicting the horizontal layout of a building. These plans include the distribution of rooms, the positioning of walls, and the placement of doors and windows, serving as essential tools for construction and spatial planning. A floor plan serves as a foundational representation of a building's layout from an aerial perspective, encompassing a horizontal projection and a corresponding legend. It is a fundamental example of a construction drawing, illustrating the shape, size, and arrangement of the structure while detailing the dimensions and materials of walls and columns, as well as the types and placements of windows and doors. This blueprint is a crucial reference for delineating lines, erecting walls, installing doors and windows, executing interior and exterior finishes, and formulating budgets during the construction phase. As shown in Figure 1, the blueprint shows a building floor plan with bathrooms, doors, windows, etc.

A floor plan is a two-dimensional (2D) horizontal projection of a building's floors, effectively conveying the layout of its spatial components, such as regions, doors, and walls.

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Review

基于3ST特征的建筑平面图检索技术综述

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摘要：近年来，建筑平面图的特征检索技术因其在施工项目高效管理与实施中的关键作用而备受关注。本文系统性地探讨了建筑平面图检索所必需的四大类特征：语义特征、空间特征、形状特征及纹理特征（统称为3ST特征）。针对这些特征，本文深入分析了其提取算法及基本原理，并重点介绍了小波变换和傅里叶形状描述子等先进方法。此外，通过严格的实验分析，对多种检索算法的性能进行了评估，为优化建筑平面图的检索提供了有价值的参考。最后，本研究还展望了平面图特征检索技术未来发展的若干方向。

关键词：平面图特征检索；3ST特征；小波变换；傅里叶形状描述

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1. 引言

在建筑设计领域，建筑蓝图与平面图是两个基本概念，然而二者的含义与功能常常被混淆。建筑蓝图涵盖了建筑工程的完整设计文件，通常包括平面图、立面图、剖面图以及其他详细图纸，用以说明建筑的结构、电气和给排水系统等。相比之下，平面图只是建筑蓝图中的一个特定组成部分，主要展示建筑物的水平布局。这些图纸包含房间的分布、墙体的位置以及门窗的布置，是施工与空间规划的重要工具。平面图是从俯视角度对建筑布局的底层表达，包含水平投影及其相应的图例，是建筑施工图的基本形式之一，用于展示建筑的形状、尺寸及空间组合，并详细标注墙体、柱子的尺寸与材料，以及门窗的类型与位置。该蓝图在施工阶段是划线定位、砌筑墙体、安装门窗、实施内外装修以及编制预算的重要依据。如图1所示，该蓝图呈现了一栋建筑的平面布局，其中包括卫生间、门、窗等要素。

平面图是建筑物各楼层的二维（2D）水平投影，能够有效展示其空间构成要素的布局，如区域、门和墙体等。

The automatic retrieval of floor plans has been actively studied over the past few decades [1]. A fundamental task in this area is segmenting a floor plan into regions (e.g., bedroom, living room) with accurate labels. However, the heterogeneous information present in floor plans complicates semantic segmentation, including tasks like line detection and region-growing segmentation [2]. With the emergence of deep learning neural networks (DNNs), DNN-based planar graph retrieval has gained significant popularity [3]. In particular, the integration of convolutional neural networks (CNNs) and recurrent neural networks (RNNs) has proven to be a powerful approach. Notable DNN architectures include fully convolutional networks (FCNs) [4], U-Net [5], and DeepLab [6]. These architectures have applications in various domains, including digitizing residential structures [7], analyzing and identifying elements in 2D building plans, constructing 3D models of buildings [8–11], evaluating the appeal of a building's layout to users [12,13], and simulating immersive virtual reality (VR) indoor architectural environments [14]. Our analysis of numerous existing studies revealed that AI holds significant potential in the architectural design process, encompassing idea development, data analysis (e.g., project predictive analysis), construction supervision, and ongoing facility maintenance. To evaluate the role of AI in the design process, we compared the latest large-scale models—DALL-E, Midjourney, and Stable Diffusion—across various design stages [15]. The study demonstrated that DALL-E and Stable Diffusion outperform Midjourney in generating ideas, sketches, and architectural style variants, significantly accelerating concept development. Stable Diffusion excelled in accurately representing construction plans and interior/exterior designs, with DALL-E closely following in performance. DALL-E also effectively addresses diverse ideation needs and offers robust editing capabilities. While Midjourney lacks features like in/out-painting and image combination, it remains valuable for basic sketching. Stable Diffusion achieves a balance, excelling in both generative design and detailed construction planning [16].

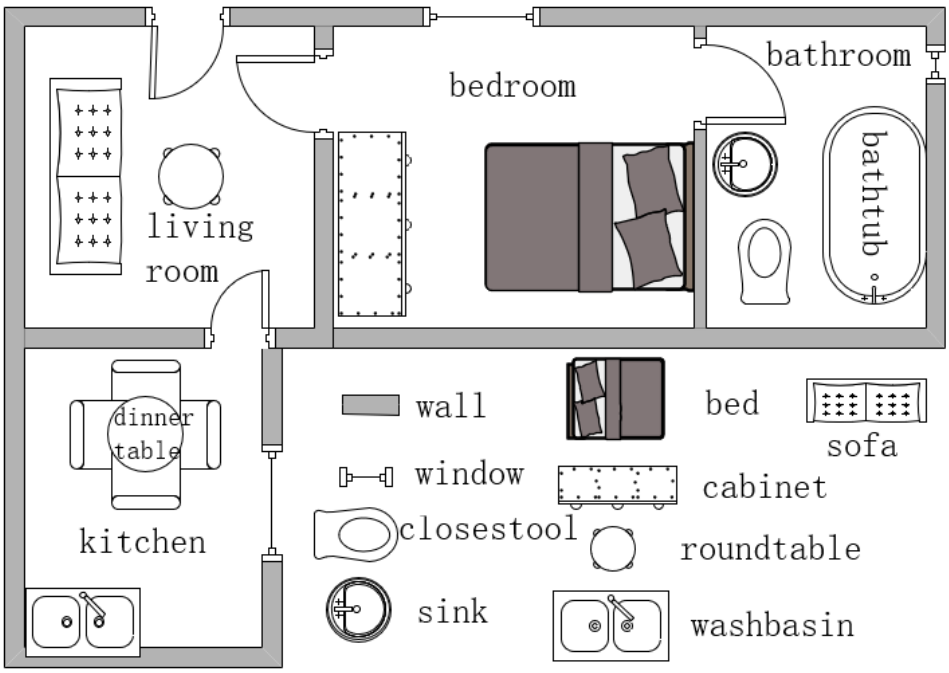


Figure 1. Building floor plan.

The complexity and size of building plans necessitate large datasets for training deep learning models; however, issues such as blurriness, overlap, and distortion can adversely affect the accuracy of retrieval models. In contrast, prompt and accurate retrieval of rele-

过去几十年来，平面图的自动提取一直是研究的热点领域[1]。该领域的基础性任务是将平面图分割为具有准确标签的区域（如卧室、客厅）。然而，平面图中存在异质性信息，这使得语义分割任务变得复杂，例如线段检测和区域生长分割等[2]。随着深度学习神经网络（DNN）的兴起，基于DNN的平面图识别技术得到了广泛应用[3]。特别是卷积神经网络（CNN）与循环神经网络（RNN）的融合已被证明是一种高效的方法。典型的DNN架构包括全卷积网络（FCN）[4]、U-Net[5]以及DeepLab[6]。这些架构在多个领域均有应用，包括住宅建筑的数字化[7]、二维建筑平面图中各要素的分析与识别、建筑物三维模型的构建[8–11]、评估建筑布局对用户的吸引力[12,13]，以及沉浸式虚拟现实（VR）室内建筑环境的模拟[14]。通过对大量现有研究的分析，我们发现人工智能在建筑设计过程中具有巨大潜力，涵盖创意构思、数据分析（如项目预测性分析）、施工监管以及设施的持续维护等环节。为评估人工智能在设计流程中的作用，我们比较了最新一批大规模模型——DALL-E、Midjourney和Stable Diffusion——在不同设计阶段的表现[15]。研究表明，在创意生成、草图绘制及建筑风格变体的生成方面，DALL-E和Stable Diffusion均优于Midjourney，显著加速了概念开发过程。Stable Diffusion在建筑施工图及室内外设计的精确表达方面表现尤为突出，DALL-E紧随其后。此外，DALL-E能够较好地满足多样化的创意需求，并具备强大的编辑能力。尽管Midjourney缺乏内/外绘画及图像组合等功能，但在基础草图绘制方面仍具有一定价值。Stable Diffusion则实现了较好的平衡，既擅长生成式设计，又能在详细的施工规划中发挥优势[16]。

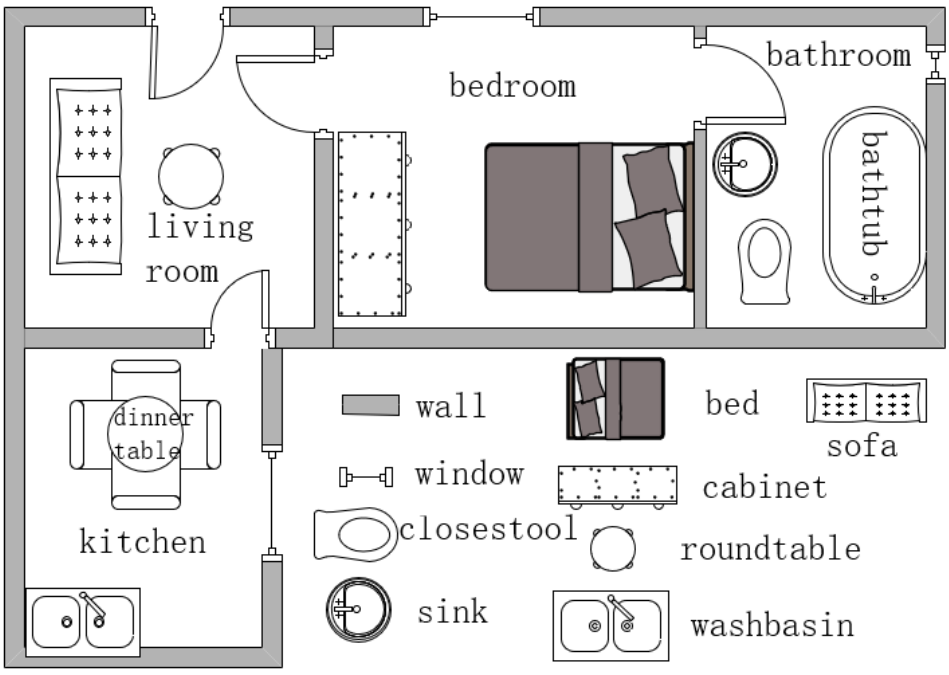


图1. 建筑平面图。

建筑图纸的复杂性和规模决定了训练深度学习模型需要庞大的数据集；然而，模糊、重叠和畸变等问题可能会对检索模型的准确性产生不利影响。相比之下，及时且准确地检索相关

vant plans can significantly enhance the efficiency of designers, engineers, and architects. A robust retrieval system enables designers to quickly identify similar design cases, fostering innovation and supporting informed design decisions. Effective management of spatial layouts, structural designs, equipment configurations, and other resources can optimize the building design and construction process, ultimately minimizing resource waste. With the advancements in building information modeling (BIM) and intelligent building technologies, the need for effective retrieval and analysis of building plan features has become increasingly vital. The advanced integration of BIM and VR in the construction environment, exemplified by immersive virtual reality-based construction management schedules, demonstrates their transformative potential. BIM offers significant advantages in controlling life cycle costs and addressing environmental challenges, enhancing construction efficiency, improving design quality, and facilitating decision-making. Studies have demonstrated that VR technology enhances stakeholder participation by 62% and spatial awareness by 48%, fostering greater community engagement and inclusiveness in the development process. The integration of BIM and VR not only optimizes construction workflows but also enhances environmental and socio-economic outcomes, including a 20% reduction in greenhouse gas emissions [16]. This paper aims to comprehensively investigate key aspects of building plan feature retrieval technology, including semantic, spatial, texture, and shape characteristics, while emphasizing their significance and potential applications.

The main contributions of this paper are as follows:

1. Through the systematic classification of various features in building floor plans, this paper provides a comprehensive framework to help researchers and practitioners understand and apply these features more effectively. This classification not only supports theoretical research but also offers valuable guidance for practical implementations.
2. Through a detailed analysis of the four features, this paper introduces innovative tools and methodologies that assist architectural designers and planners in selecting and optimizing design schemes. It illustrates how these tools work together to extract and analyze features of building floor plans, contributing to improved design efficiency and effectiveness.

2. Floor Plan Retrieval Overview

2.1. Overview of Floor Plan Feature Extraction

The floor plan retrieval method based on deep learning primarily employs models such as CNNs, RC-Net, Faster R-CNN, Transformers, and Graph Matching Networks (GMNs) to extract features from building floor plans. By identifying unique high-level semantic features and employing feature vectors to calculate image similarity, this approach enables effective retrieval of building floor plans. Prior to 2013, the forward extraction method was predominantly utilized, directly extracting relevant information or features from images, such as identifying and isolating structural components like walls, doors, and windows [17,18]. However, the efficiency of direct recognition is limited due to the lack of distinct features for these elements in the floor plan, whereas functional components, such as beds and tables, are more clearly defined [19]. To address this challenge, the reverse extraction method was introduced in 2019, which derives or generates new information from previously extracted data and utilizes algorithmic modeling to identify and remove functional components, thereby enhancing extraction efficiency [20].

In 2022, Shehzadi et al. [21] utilized a CNN to identify connection points in building floor plans, such as wall corners and door endpoints. Their method established links between these intersections and implemented an integer programming algorithm to detect doors on walls. In 2021, Fan et al. [22] proposed a novel approach that integrated graph convolutional networks (GCNs) with CNNs in an end-to-end architecture. This architecture

建筑平面方案能够显著提升设计师、工程师和建筑师的工作效率。一个稳健的检索系统使设计人员能够快速识别相似的设计案例，从而促进创新并支持基于充分信息的设计决策。对空间布局、结构设计、设备配置及其他资源的有效管理，可优化建筑设计与施工流程，最终减少资源浪费。随着建筑信息模型（BIM）与智能建造技术的不断发展，对建筑平面特征进行有效检索与分析的需求日益迫切。BIM与虚拟现实（VR）在施工环境中的深度融合，例如基于沉浸式虚拟现实的施工进度管理，充分展现了其变革性潜力。BIM在控制全生命周期成本、应对环境挑战、提升施工效率、改善设计质量以及辅助决策等方面具有显著优势。研究表明，VR技术可使利益相关者的参与度提高62%，空间感知能力提升48%，从而促进社区在开发过程中的更广泛参与与包容性。BIM与VR的集成不仅优化了施工工作流，还提升了环境与社会经济绩效，包括将温室气体排放量降低20%[16]。本文旨在全面探讨建筑平面特征检索技术的关键方面，包括语义特征、空间特征、纹理特征及形状特征，并着重阐述其重要性与潜在应用。本文的主要贡献如下：

1. 通过对建筑平面图中各类特征的系统分类，本文提供了一个全面的框架，有助于研究人员和从业人员更有效地理解和应用这些特征。该分类不仅支持理论研究，也为实际应用提供了有价值的指导。 2. 通过针对四项特征的详细分析，本文提出了创新的工具与方法，协助建筑设计师和规划师进行设计方案的遴选与优化。文中阐述了这些工具如何协同工作，以提取并分析建筑平面图的特征，从而提升设计效率与效果。

2. 平面图检索概述

2.1. Overview of Floor Plan Feature Extraction

基于深度学习的建筑平面图检索方法主要采用卷积神经网络（CNN）、RC-Net、Faster R-CNN、Transformer以及图匹配网络（GMN）等模型，从建筑平面图中提取特征。通过识别独特的高层语义特征，并利用特征向量计算图像相似度，该方法能够实现建筑平面图的有效检索。在2013年之前，主要采用正向提取方法，即直接从图像中提取相关信息或特征，例如识别并分离墙体、门窗等结构构件[17,18]。然而，由于建筑平面图中这些结构要素的特征不够显著，直接识别的效率受到限制；相比之下，床、桌等功能构件的特征则更为清晰[19]。为应对这一挑战，2019年提出了逆向提取方法，该方法从已提取的数据中推导或生成新的信息，并借助算法建模识别并去除功能构件，从而提升提取效率[20]。

2022年，Shehzadi等人[21]利用卷积神经网络（CNN）识别建筑平面图中的连接点，如墙角和门端点。他们的方法在这些交点之间建立关联，并采用整数规划算法来检测墙上的门。2021年，Fan等人[22]提出了一种新颖的方法，该方法在端到端架构中将图卷积网络（GCN）与卷积神经网络相结合。该架构

comprises a CNN backbone, a graph convolution header, and a detection header, collectively forming a baseline network for panoramic symbol recognition tasks. The CNN-GCN approach achieved state-of-the-art (SOTA) performance in semantic symbol recognition. In 2023, Wang et al. [23] employed the RC-Net framework to extract fundamental features from floor plan images using a VGG encoder. Their feature branch emphasized room boundaries and types, optimizing the learning of essential features necessary for prediction. The final output is a room mask that integrates text features from the text branch with room features to enhance accuracy. These studies indicate that the advancement of Graph Neural Networks (GNNs) has led to floor plan retrieval based on graph structures, demonstrating that GNNs can outperform Faster R-CNN in terms of performance and accuracy, making them well suited for handling large-scale architectural floor plans [24]. In recent years, scholars [7] have begun exploring techniques for converting 2D floor plans into 3D models by first segmenting structural components, such as walls, and recognizing associated symbols, including windows and doors. They subsequently extract wall details from the predicted segmentation mask to generate semantic elements, which can then be utilized to construct 3D models in accordance with the Industrial Foundation Class (IFC) standard.

Regarding the conversion of 3D models into building floor plans, a spatial context module has been proposed to enhance the accuracy of room type prediction by transferring room boundary features from the top-layer decoder to the bottom-layer decoder [25]. Additionally, deep segmentation and detection neural networks have been employed to extract structural information from rooms, determine room dimensions through keypoint detection and clustering analysis, vectorize the room data via an iterative optimization method, and generate vectorized 3D reconstruction outputs [26]. Furthermore, three augmentation techniques—Gaussian noise, Gaussian blurring, and random rotation—have been utilized to enhance the input floor plans [27]. Multiple rounds of random erosion operations are performed on the target wall images to decrease the model's sensitivity to pixel variations at the wall edges, thereby improving its ability to effectively learn the features of wall edges. In summary, floor plan retrieval relies on image processing, feature extraction, image matching, and machine learning, involving a wide range of applications and complex algorithms. As data continue to grow, new techniques are constantly being developed and refined.

2.2. Overview of Floor Plan Retrieval Architecture

2.2.1. Network Feedforward Solutions

In the field of architecture, extracting deep features from building floor plans is essential for tasks such as building identification, design, and analysis. The emergence of deep learning techniques has significantly advanced computer vision and image processing, leading to the development of innovative feedforward neural network architectures specifically tailored for extracting complex features from building floor plans. In ref. [25], a novel approach is introduced for recognizing elements within building floor plans through the construction of a hierarchical model of floor plan components. This method employs the VGG deep convolutional neural network architecture in conjunction with a room boundary-guided attention mechanism to enhance floor plan recognition. By predicting both room boundary elements and room types, the model organizes floor plan components into a hierarchical structure, classifying them based on their placement within the interior or exterior of the building. Interior elements are further categorized into room boundary components (including walls, doors, and windows) and room type components (such as living rooms, bathrooms, and bedrooms), as shown in Figure 2, the VGG deep convolutional network can be used to obtain the features of various elements and boundary components from Figure 2.

该模型由一个CNN主干网络、一个图卷积头部和一个检测头部组成，共同构成用于全景符号识别任务的基线网络。基于CNN与GCN的联合方法在语义符号识别任务中取得了当前最优（SOTA）性能。2023年，Wang等[23]采用RC-Net框架，利用VGG编码器从平面图图像中提取基础特征；其特征分支着重于房间边界与类型信息，从而优化了预测所需关键特征的学习过程。最终输出为一张房间掩码，通过融合文本分支中的文本特征与房间特征以提升预测精度。这些研究表明，图神经网络（GNN）的发展已推动了基于图结构的平面图检索技术的进步；实验结果表明，GNN在性能与精度方面可优于Faster R-CNN，因而特别适用于处理大规模建筑平面图[24]。近年来，有学者[7]开始探索将二维平面图转换为三维模型的技术：首先对墙体等结构构件进行分割，并识别窗、门等关联符号；随后从预测的分割掩码中提取墙体细节，生成语义要素，进而依据工业基础类（IFC）标准构建三维模型。

在将三维模型转换为建筑平面图方面，已有研究提出了一种空间上下文模块，通过将顶层解码器中的房间边界特征传递至底层解码器，以提升房间类型预测的准确性[25]。此外，还采用深度分割与检测神经网络从房间中提取结构信息，借助关键点检测与聚类分析确定房间尺寸，并通过迭代优化方法对房间数据进行矢量化处理，进而生成矢量化三维重建结果[26]。进一步地，研究者利用高斯噪声、高斯模糊和随机旋转三种数据增强技术对输入的平面图进行增强[27]。通过对目标墙体图像实施多轮随机腐蚀操作，降低了模型对墙体边缘像素变化的敏感性，从而提升了其对墙体边缘特征的有效学习能力。综上所述，建筑平面图检索依赖于图像处理、特征提取、图像匹配及机器学习等技术，涉及广泛的应用场景与复杂的算法体系。随着数据规模的持续增长，相关技术也在不断演进与完善。

2.2. Overview of Floor Plan Retrieval Architecture

2.2.1 网络前馈解决方案

在建筑领域，从建筑平面图中提取深层特征对于建筑识别、设计与分析等任务至关重要。深度学习技术的兴起极大地推动了计算机视觉与图像处理的发展，促使一系列专为从建筑平面图中提取复杂特征而设计的前馈神经网络架构得以问世。文献[25]提出了一种新颖的方法，通过构建建筑平面图组件的层次化模型来识别其中的各类元素。该方法采用VGG深度卷积神经网络架构，并结合基于房间边界引导的注意力机制，以提升平面图识别性能。通过同时预测房间边界元素与房间类型，该模型将平面图组件组织成一个层次结构，并依据其在建筑内部或外部的位置进行分类。其中，内部元素进一步细分为房间边界组件（包括墙体、门和窗）以及房间类型组件（如客厅、卫生间和卧室），如图2所示；利用VGG深度卷积网络可从图2中获取各类元素及边界组件的特征表示。

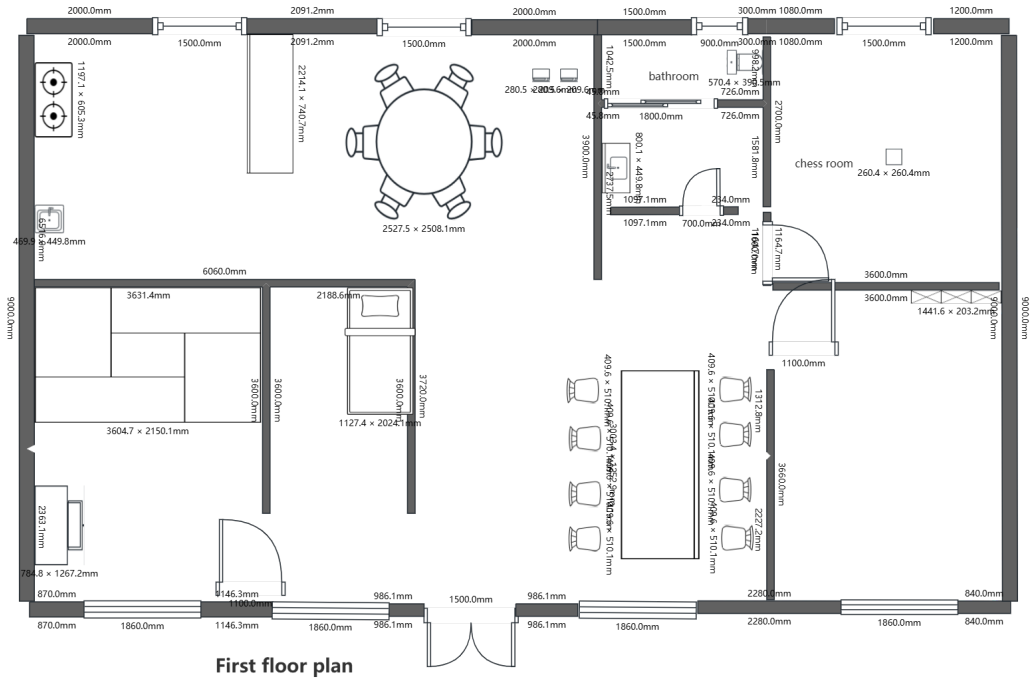


Figure 2. Building identification hierarchy diagram.

Ouahbi et al. [28] introduced a feature extraction framework. This framework consists of a contraction path with multiple convolutional and pooling layers, as well as a symmetric expansion path that includes an upsampling layer (transposed convolution) and regular convolution operations. The U-Net model effectively extracts features by merging shallow and deep information, capturing both local and global context within building floor plans. The upsampling layer facilitates feature fusion and spatial recovery, while the contraction path progressively consolidates low-level features into more abstract representations with larger receptive fields. The expansion path then restores these features to super-resolution, ensuring dense segmentation at the original resolution.

To enhance planar segmentation and vectorization processes, researchers have proposed the DeepLabv3+ network as the foundational model. By integrating dilated convolution and spatial pyramid pooling techniques, this framework excels at extracting multi-scale features, including structural elements, textual annotations, and symbolic representations from pixel maps. This advanced approach accurately identifies room configurations, dimensions, and types, effectively capturing the intricate structure and detailed nuances of buildings across various scales. The system significantly improves accuracy and generalization capabilities, particularly excelling in planar segmentation and vectorization tasks, including the detection of slant walls. The degree of match between the optimized polygons and the room contours is represented by $\iota_{boundary}$, while the match between the polygonal regions and the internal areas of the room is denoted by ι_{IOU}.

$$\iota_{boundary} = \sum_{b \in B} \min_{p_i \in P} D(p_i p_{i+1}, b) \tag{1}$$

$$\iota_{IOU} = IOU(Rasterized(P), c) \tag{2}$$

In Equation (1), $D(p_i p_{i+1}, b)$ denotes the shortest distance from b to line segment $p_i p_{i+1}$, and $Rasterized(P)$ in Equation (2) denotes the region covered by polygonal rasterization.

A groundbreaking study referenced in [29] introduces a novel neural network architecture aimed at understanding both layout types and room types within architectural blueprints. This sophisticated architecture incorporates an n-dimensional fully connected

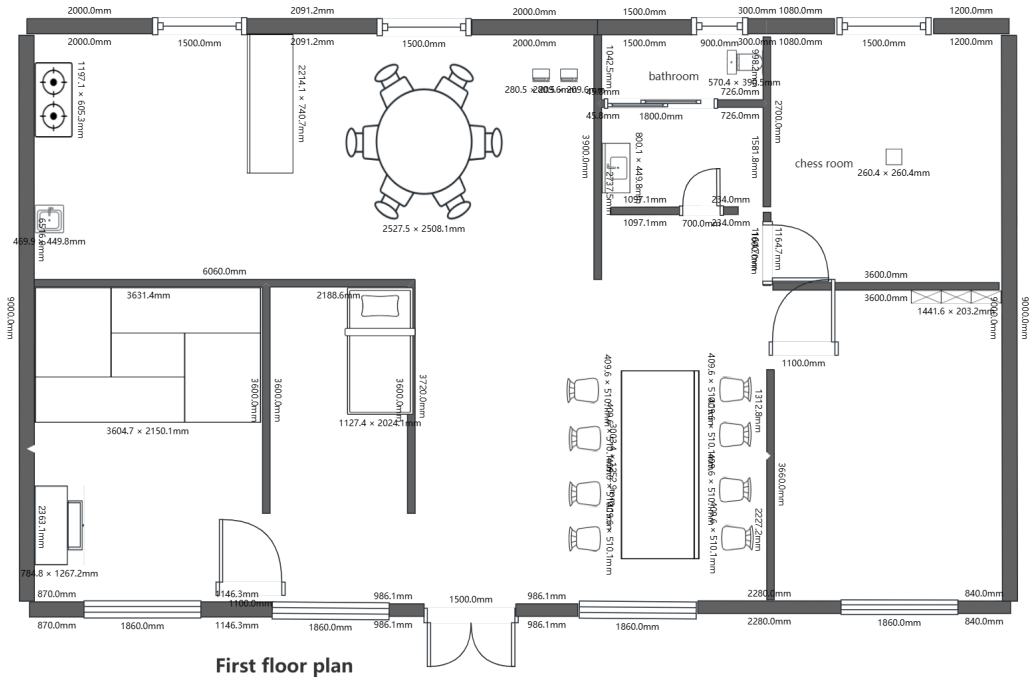


图2. 建筑物识别层级示意图。

Ouahbi 等人[28]提出了一种特征提取框架。该框架由包含多层卷积和池化操作的收缩路径，以及包含上采样层（转置卷积）和常规卷积操作的对称扩张路径组成。U-Net 模型通过融合浅层与深层信息，有效提取特征，从而在建筑平面图中同时捕捉局部与全局上下文。上采样层有助于特征融合与空间重建，而收缩路径则逐步将低级特征整合为具有更大感受野的更抽象表示。随后，扩张路径将这些特征恢复至超分辨率，以确保在原始分辨率下实现稠密分割。

为提升平面分割与矢量化处理的效果，研究人员提出了以DeepLabv3-v1网络为基础的模型。该框架通过融合空洞卷积与空间金字塔池化技术，能够高效提取多尺度特征，包括结构要素、文字标注及符号表达等像素级信息。这一先进方法可准确识别房间的布局、尺寸与类型，有效捕捉建筑在不同尺度下的复杂结构与细节特征。该系统显著提升了精度与泛化能力，在平面分割与矢量化任务中表现尤为突出，尤其擅长斜墙的检测。优化后的多边形与房间轮廓之间的匹配程度用v2表示，而多边形区域与房间内部区域之间的匹配程度则用v3表示。

$$\iota_{boundary} = \sum_{b \in B} \min_{p_i \in P} D(p_i p_{i+1}, b) \tag{1}$$

$$\iota_{IOU} = IOU(Rasterized(P), c) \tag{2}$$

在式（1）中，$D(p_i p_{i+1}$、$b)$表示点b到线段$p_i p_{i+1}$的最短距离；而在式（2）中，$Rasterized(P)$表示由多边形光栅化所覆盖的区域。

一项具有开创性意义的研究[29]提出了一种新颖的神经网络架构，旨在同时识别建筑蓝图中的布局类型与房间类型。该架构采用了一个n维全连接层。

layer placed behind the fc7 layer of the widely used VGG-16 model [30], which has been pre-trained on the extensive ImageNet dataset. In addition, a 2 m-dimensional fully connected (FC) layer is specifically designed to classify various floor types, while a parallel set of 2 m-dimensional FC layers collaborates to identify distinct room types. Here, n represents the different layout types under investigation, while m denotes the room types awaiting classification.

Our analysis of the parameter optimization process in VGG-16 revealed a significant limitation: the process is governed by a feedback loop between layout-type classification errors and room existence classification errors. While parameter optimization in VGG-16 allows for the fine-tuning of the network's parameters, it neglects the critical role of edges within the complex network of building information. Consequently, the network fails to capture the intricate details embedded in the interconnections between rooms, which limits its capacity to comprehend the spatial relationships inherent in the building layout.

2.2.2. Feature Extraction of Floor Plans Structural Elements

In building plan analysis, the selection of deep features emphasizes extracting relevant insights from raw blueprints to enhance operational efficiency. When extracting wall features, the inherent variability and irregularity of their shapes and sizes necessitate sophisticated approaches. For instance, the ResNet-based Feature Pyramid Network (FPN) architecture is integrated with semantic segmentation techniques. In contrast, for features such as doors and windows, which are characterized by discrete and repetitive symbols, the bounding box detection neural network is seamlessly combined with the Faster R-CNN model, yielding effective results.

Since room-type text can provide valuable insights into the functionality of a specific space, Wang et al. [23] employed a text branching technique to extract textual features and integrated them with spatial features through a merging module. This approach was designed to enhance semantic features for subsequent predictive analysis. Furthermore, Gao et al. [27] conducted a comprehensive statistical analysis to assess the significance of features during the deep feature selection process. Their methodology involved segmenting the drawing box, extracting wall features by isolating the wall centerline, and feeding the processed data into a space-merging module for boundary completion and integration. The resulting data were meticulously filtered to eliminate errors, ensuring the accuracy and reliability of the final output. In these studies, segmentation networks were primarily utilized to extract key structural elements, such as walls, rooms, windows, and doors. Vectorization algorithms were subsequently applied to these elements, and detection models were employed to identify symbols and text within the extracted data. This information, combined with the length of measured lines, facilitated the calculation of the floor plan's scale.

2.2.3. Similarity Measure

YAMASAKI et al. [18] introduced a novel approach for planar graph retrieval based on a fully convolutional network (FCN) and a graph model structure. This method applies a semantic segmentation algorithm to planar graphs using the FCN to categorize each component. Vertices are established when the segmented object exceeds a threshold size of 1000, and edges are added if the distance between objects is less than 30, transforming the planar graph into a graph model structure. Additionally, a new algorithm is devised and optimized to extract MCS [31] and isolated points, with similarity calculations on MCS used for retrieving images with similar structures.

$$s(v) = \mathrm{sim}(s_g, s_h) = \frac{min(s_g, s_h)}{max(s_g, s_h)} \quad (3)$$

该层被置于广泛使用的VGG-16模型[30]的fc7层之后，该模型已在规模庞大的ImageNet数据集上进行了预训练。此外，一个2m维的全连接（FC）层专门用于对各类地面类型进行分类，而一组并行的2m维全连接层则协同完成对不同房间类型的识别。其中，n表示待研究的不同布局类型，m则表示待分类的房间类型。

我们对VGG-16中参数优化过程的分析揭示了一个显著的局限性：该过程受布局类型分类错误与房间存在性分类错误之间反馈环路的主导。尽管VGG-16中的参数优化能够对网络参数进行微调，却忽视了建筑信息复杂网络中边缘所发挥的关键作用。因此，该网络无法捕捉房间间相互连接所蕴含的精细细节，从而限制了其对建筑布局中固有空间关系的理解能力。

2.2.2. 平面图结构构件的特征提取

在建筑平面图分析中，深度特征的选取侧重于从原始蓝图中提取相关洞察，以提升运行效率。在提取墙体特征时，其形状与尺寸的固有多样性和不规则性 necessitate 采用更为复杂的方法。例如，基于ResNet的特征金字塔网络（FPN）架构被整合到语义分割技术中。相比之下，对于门和窗等具有离散且重复性符号的特征，则将边界框检测神经网络与Faster R-CNN模型无缝结合，从而取得良好的效果。

由于房间类型的文本能够提供关于特定空间功能的宝贵信息，Wang 等人[23]采用文本分支技术提取文本特征，并通过一个融合模块将其与空间特征相结合，旨在增强语义特征以支持后续的预测分析。此外，Gao 等人[27]在深度特征选择过程中开展了全面的统计分析，以评估各特征的重要性。他们的方法包括对图纸边界框进行分割，通过提取墙体中心线来获取墙体特征，并将处理后的数据输入到空间融合模块中，以完成边界补全与特征整合。随后，对所得数据进行细致的过滤，以消除误差，确保最终结果的准确性和可靠性。在这些研究中，主要利用分割网络提取关键结构要素，如墙体、房间、窗户和门；随后对这些要素应用矢量化算法，并借助检测模型识别提取数据中的符号与文本。结合测量线段的长度，上述信息可用于计算平面图的比例尺。

2.2.3 相似性度量

YAMASAKI 等人[18]提出了一种基于全卷积网络（FCN）和图模型结构的平面图检索新方法。该方法利用 FCN 对平面图实施语义分割，以对各组件进行分类。当分割得到的物体面积超过 1000 像素时即建立顶点；若两物体之间的距离小于 30 像素，则添加边，从而将平面图转换为图模型结构。此外，还设计并优化了一种新算法，用于提取最大共同子图（MCS）及孤立点，并通过对 MCS 进行相似性计算来检索具有相似结构的图像。

$$s(v) = \mathrm{sim}(s_g, s_h) = \frac{min(s_g, s_h)}{max(s_g, s_h)} \tag{3}$$

$$w(e) = 2e^{s(v_{e1})} * e^{s(v_{e2})} \tag{4}$$

In Equation (3), $s(v)$ is denoted as the similarity of vertices s_g and s_h in graphs G and H. It is obtained by comparing the size of the set of vertices in the two graphs using Equation (4), where $w(e)$ is the edge weight between two vertices v_{e1} and v_{e1}. The similarity between graphs G and H is ultimately computed by summation:

$$sim(G,H) = \sum_{ein\ M} w(e) + \sum_{isolatevinein\ M} s(v) \tag{5}$$

For multimodal feature extraction, YUKI et al. [29] proposed a method to transform planar images into graph structures. The authors contend that the variability in drawing styles across floor plans adversely affects the performance of conventional neural networks. To address this challenge, they introduced a framework that jointly optimizes room layout and room style classifiers. Features were extracted using the VGG network, which transforms the structural characteristics of rooms into graph-based representations, thereby enabling graph-based retrieval of similar attributes in floor plans. This framework achieved an accuracy of 0.49 at $p = 0.5$. DIVYA et al. [32] proposed a unified framework that employs a graph embedding method to represent graphs extracted from layouts. A two-stage matching and retrieval method, consisting of Room Layout Matching (RLM) and Room Decoration Matching (RDM), was proposed for feature matching and retrieval. The evaluation primarily focuses on ranking retrieval results to assess the method's capability to identify layouts similar to the query within the database. However, this approach exhibits several limitations: the two-stage matching process lacks uniformity; critical features such as room dimensions, which are essential for buyers, are overlooked; and the dataset is constrained by limited sample variation. To address these challenges, DIVYA et al. [33] introduced an end-to-end framework for extracting high-level semantic features, such as room sizes, adjacencies, and furnishings, within a fine-grained retrieval approach. This method utilizes feature fusion to aggregate high-level semantic features, enabling the retrieval of similar building floor plans. The total matching score M is computed by weighting the four extracted feature scores.

$$M(i,j) = \frac{\rho^+(i,j) + \psi^+(i,j) + \phi^+(i,j) + \theta^+(i,j)}{4} \tag{6}$$

In Equation (6), ρ^+ delineates the RAS (Room Adjacency String) score, ψ^+ denotes the CAR (Ratio of Carpet Area) score, ϕ^+ indicates the number of furniture score, and θ^+ indicates the type of furniture score.

In addition to the above similarity calculations, there are actually L_1 and L_2 distances, Mahalanobis distances, and histogram intersections. If the components of the image features are orthogonal and independent, and the importance of each dimension is the same, the distance between two feature vectors A and B can be measured by L_1 distance or L_1 distance (also known as Euler distance). The L_1 distance can be expressed as

$$D_1 = \sum_{i=1}^{N} |A_i - B_i|, \tag{7}$$

where N is the dimension of the feature vector. Similarly, the L_2 distance can be expressed as

$$D_2 = \sum_{i=1}^{N} (A_i - B_i)^2. \tag{8}$$

$$w(e) = 2e^{s(v_{e1})} * e^{s(v_{e2})} \tag{4}$$

在式（3）中，$s(v)$ 表示图 G 和图 H 中顶点 s_g 与 s_h 的相似度。该相似度通过比较两图中顶点集的大小获得，具体计算见式（4），其中 $w(e)$ 为顶点 v_{e1} 与 v_{e1} 之间的边权重。图 G 与图 H 之间的相似度最终通过求和得到：

$$sim(G,H) = \sum_{ein\ M} w(e) + \sum_{isolatevinein\ M} s(v) \tag{5}$$

在多模态特征提取方面，YUKI 等人[29]提出了一种将平面图转换为图结构的方法。作者认为，不同楼层平面图中绘图风格的多样性会显著影响传统神经网络的性能。为此，他们设计了一个联合优化房间布局与房间风格分类器的框架。该框架采用 VGG 网络提取特征，将房间的结构特征转化为基于图的表示形式，从而实现对楼层平面图中相似属性的图级检索。在阈值 v12=0.5 下，该框架的准确率达到 0.49。DIVYA 等人[32]则提出了一种统一框架，利用图嵌入方法对从布局中提取的图进行表示，并进一步提出了由“房间布局匹配”（RLM）和“房间装饰匹配”（RDM）组成的两阶段匹配与检索方法，用于特征匹配与检索。其评估主要聚焦于检索结果的排序，以衡量该方法在数据库中识别与查询相似布局的能力。然而，该方法存在若干局限：两阶段匹配过程缺乏一致性；对于买家至关重要的房间尺寸等关键特征被忽略；且数据集的样本多样性有限。针对上述问题，DIVYA 等人[33]进一步提出了一种端到端框架，在细粒度检索范式下提取房间尺寸、邻接关系及家具配置等高层语义特征。该方法通过特征融合聚合高层语义特征，从而实现相似建筑楼层平面图的检索。最终匹配得分 v13 由四个提取特征得分的加权求和得到。

$$M(i,j) = \frac{\rho^+(i,j) + \psi^+(i,j) + \phi^+(i,j) + \theta^+(i,j)}{4} \tag{6}$$

在式（6）中，ρ^+ 表示 RAS（房间邻接字符串）得分，ψ^+ 表示 CAR（地毯面积比）得分，ϕ^+ 表示家具数量得分，θ^+ 表示家具类型得分。

除了上述的相似性计算方法外，实际上还有L_1距离、L_2距离、马氏距离以及直方图交集等。若图像特征的各分量彼此正交且相互独立，且各维度的重要性相同，则可通过L_1距离或L_1距离（亦称欧几里得距离）来度量两个特征向量A与B之间的距离。L_1距离可表示为

$$D_1 = \sum_{i=1}^{N} |A_i - B_i|, \tag{7}$$

其中，N 为特征向量的维度。同理，L_2 距离可表示为

$$D_2 = \sum_{i=1}^{N} (A_i - B_i)^2. \tag{8}$$

If the components of the feature vector are correlated or have different weights, the Mahalanobis distance can be used to calculate the similarity between the features. The mathematical expression of the Mahalanobis distance is

$$D_{\text{mahal}} = (A - B)^t C^{-1} (A - B), \tag{9}$$

where C represents the covariance matrix of the feature vector. This distance criterion is frequently employed to assess the similarity of Synthetic Aperture Radar (SAR) features.

2.3. Overview of 3ST Feature Analysis

Feature extraction is a fundamental concept in computer vision and image processing. It refers to the process of using computers to extract information from images and determine whether each image point corresponds to a specific image feature. The outcome of feature extraction is the division of image points into distinct subsets, which typically comprise isolated points, continuous curves, or continuous regions. As of now, there is no universally accepted and precise definition of features. Various image features and similarity algorithms are employed in image retrieval. For a specific image library, it is essential to select one or more of the most effective image features and similarity algorithms. This necessitates a comprehensive evaluation of the retrieval effectiveness under varying conditions, comparing the advantages and disadvantages of diverse methods to identify the most optimal approach.

Semantic feature extraction is a fundamental operation in image processing, typically performed as the initial step in analyzing an image. It involves examining each pixel to identify whether it corresponds to a feature. As feature extraction serves as a critical computational step in numerous image processing algorithms, a wide range of feature extraction methods have been developed, each differing in the types of features extracted, computational complexity, and repeatability. An edge, for instance, is a pixel that delineates the boundary between two distinct image regions. The shape of an edge can vary arbitrarily and may include intersections. In practice, edges are defined as subsets of image points exhibiting significant gradients. Common algorithms often connect points with high gradients to construct a more comprehensive representation of the edge. These algorithms may impose constraints on the edge, which is locally a one-dimensional structure. Corners are point-like features in an image, characterized by a two-dimensional structure locally. Early algorithms initially performed edge detection and subsequently analyzed edge directions to identify abrupt changes, corresponding to corners. Later algorithms eliminate the need for edge detection and instead directly identify regions of high curvature in the image gradient.

The performance of texture features, ranked from best to worst, is as follows: multidimensional autoregressive texture feature (MRSAR), Gabor feature, TWT wavelet transform, PWT wavelet transform, improved Tamura feature, roughness histogram, directional histogram, and traditional Tamura feature. Additionally, the performance of the traditional Tamura feature exceeds that of both the roughness and directional histograms, indicating that it is more suitable for describing image sets with relatively uniform textures, such as Brodatz.

Geometric shape features and color aggregation vectors are more effective for images with uniform colors or textures, offering no advantages for general images. Furthermore, the geometric method is a texture feature analysis approach based on the theory of texture primitives (basic texture elements). The texture primitive theory posits that complex textures can be composed of several simple texture primitives arranged in a regular, repetitive pattern.

如果特征向量的各分量之间存在相关性或具有不同的权重，则可采用马氏距离来度量特征之间的相似性。马氏距离的数学表达式为：

$$D_{\text{mahal}} = (A - B)^t C^{-1} (A - B), \tag{9}$$

其中，C 表示特征向量的协方差矩阵。该距离准则常用于评估合成孔径雷达（SAR）特征的相似性。

2.3. Overview of 3ST Feature Analysis

特征提取是计算机视觉与图像处理领域的一个基本概念。它是指利用计算机从图像中提取信息，并判断每个图像点是否对应于某一特定的图像特征的过程。特征提取的结果是将图像点划分为若干个互不重叠的子集，这些子集通常由孤立点、连续曲线或连续区域构成。截至目前，关于“特征”尚无被广泛接受且精确的定义。在图像检索中，人们使用了多种图像特征及相似性度量算法。对于特定的图像库而言，必须选择一种或多种最为有效的图像特征与相似性度量算法。为此，有必要在不同条件下对检索效果进行综合评估，比较各种方法的优劣，从而确定最优的实现方案。

语义特征提取是图像处理中的基本操作，通常作为图像分析的初始步骤。它涉及对每个像素进行检查，以确定其是否对应于某一特征。由于特征提取是众多图像处理算法中的关键计算环节，因此已发展出多种特征提取方法，这些方法在所提取的特征类型、计算复杂度以及重复性等方面各不相同。例如，边缘是指分隔两个不同图像区域的像素；边缘的形状可以任意变化，并可能包含交点。在实践中，边缘被定义为具有显著梯度的图像点的子集。常见的算法往往将高梯度点连接起来，以构建更为完整的边缘表示。这些算法可能会对边缘施加约束，使其在局部上表现为一维结构。角点则是图像中的一种点状特征，其局部结构为二维。早期的算法首先执行边缘检测，随后分析边缘方向以识别突变点，即角点。后来的算法则无需进行边缘检测，而是直接识别图像梯度中曲率较高的区域。

纹理特征的性能由优至劣依次为：多维自回归纹理特征（MRSAR）、Gabor特征、TWT小波变换、PWT小波变换、改进的Tamura特征、粗糙度直方图、方向直方图以及传统的Tamura特征。此外，传统Tamura特征的性能优于粗糙度直方图和方向直方图，这表明其更适用于描述纹理相对均匀的图像集，例如Brodatz数据集。

几何形状特征与颜色聚合向量对于具有均匀颜色或纹理的图像更为有效，而对于一般图像则不具备优势。此外，基于纹理基元（基本纹理单元）理论的几何方法是一种纹理特征分析技术。该理论认为，复杂的纹理可由若干简单的纹理基元按照规则的、重复的模式排列而成。

The use of spatial relationship features enhances the ability to describe and distinguish image content; however, these features are often sensitive to transformations such as rotation, inversion, and scale changes in images or targets. Moreover, in practical applications, relying solely on spatial information is often insufficient, as it cannot effectively or accurately represent scene information. To enable retrieval, other features, in addition to spatial relationship features, are also required. For details, see the following section.

3. Semantic Feature Retrieval

3.1. Semantic Feature Analysis

Feature extraction from building plans is challenging due to interfering elements like thin axes and walls of specific thicknesses. To mitigate the impact of these interfering lines and highlight wall features, recent research [27] employed linear downsampling on input images, resizing them to 512 × 512 pixels to approximate the receptive field size of ResNet50 (483 × 483 pixels). Additionally, two iterations of erosion with a 3 × 3 filtering kernel were applied to replace hollow walls with solid ones. To address data scarcity, three data augmentation techniques were applied to the input plan views: Gaussian noise, Gaussian blurring, and random rotation, all aimed at mitigating overfitting. Furthermore, zero to two rounds of random erosion were applied to target wall images to reduce sensitivity to pixel variations along wall edges, improving the model's ability to learn wall edge features.

To enhance the semantic information of room type features, a study [34] employed combinatorial mapping and its dual approach. The method used two VGG-based feature extraction branches to predict boundaries and room types, integrating orientation-aware kernels and boundary features to enrich semantic information and generate a compact representation of both geometric and semantic details. By leveraging machine learning algorithms, this approach showed strong potential for floor plan analysis across diverse room styles, highlighting its significant generalization capabilities.

3.2. Rule Feature Extraction Retrieval

Prior to the advent of deep learning methodologies, traditional approaches to building floor plan retrieval relied on manually crafted features (e.g., match points, histograms, and eigenvalues) [35], as well as rule-based techniques. In building floor plan retrieval, fundamental graphical components (such as rooms, doors, and windows) must first be identified within the architectural layout [36]. Conventional methods typically separate textual and image data, employing handcrafted rules to identify elements within building floor plans. However, reliance on manual feature design and rule-based formulations presents several limitations. First, manual creation of features and rules requires specialized expertise but often fails to address the complexities and variations in diverse building layouts, resulting in error-prone outcomes. Second, this approach lacks universality and fails to adapt to different architectural styles, requiring extensive customization for each scenario.

As early as 1996, a study [37] introduced a prototype system for the automated recognition and interpretation of hand-drawn architectural floor plans. The system employed pattern recognition techniques to analyze processed images, extract architectural components (e.g., rooms, walls, doors, and windows), and organize them into a structured format. In 2011, another study [38] used Speeded Up Robust Features (SURFs), including edge extraction and boundary detection, to extract localized information from architectural floor plans. The extracted structural data were then used for analysis and retrieval tasks. In content-based image retrieval (CBIR), symbolic localization enabled document retrieval based on a query image, allowing for the identification of the query image's specific location. Additionally, a graph-based retrieval method, known as the "room connectivity graph", was proposed in [39]. This method extracted room connectivity graphs from

空间关系特征的使用有助于增强对图像内容的描述与区分能力；然而，这些特征往往对图像或目标的旋转、翻转及尺度变化等变换较为敏感。此外，在实际应用中，仅依赖空间信息通常不足以有效且准确地表征场景信息。因此，为了实现检索，除了空间关系特征之外，还需引入其他类型的特征。具体内容详见下文。

3. 语义特征检索

3.1. Semantic Feature Analysis

由于细长的轴线以及特定厚度的墙体等干扰因素的存在，从建筑平面图中进行特征提取具有较大挑战。为削弱这些干扰线的影响并突出墙体特征，近期研究[27]对输入图像进行了线性下采样，将其尺寸调整为512×512像素，以近似ResNet50的感受野大小（483×483像素）。此外，还使用3×3的腐蚀滤波核对图像进行两次迭代处理，将空心墙体替换为实心墙体。为应对数据稀缺问题，对输入的平面视图采用了三种数据增强技术：高斯噪声、高斯模糊和随机旋转，旨在缓解过拟合现象。进一步地，针对墙体图像施加零至两轮的随机腐蚀操作，以降低模型对墙体边缘像素变化的敏感性，从而提升其学习墙体边缘特征的能力。

为增强房间类型特征的语义信息，一项研究[34]采用了组合映射及其对偶方法。该方法使用两个基于VGG的特征提取分支来预测边界和房间类型，并通过引入方向感知核与边界特征，丰富语义信息，从而生成兼具几何与语义细节的紧凑表示。借助机器学习算法，该方法在面向多种房间风格的平面图分析中展现出较强的潜力，凸显了其显著的泛化能力。

3.2. Rule Feature Extraction Retrieval

在深度学习方法出现之前，传统的建筑平面图检索技术主要依赖于人工设计的特征（如匹配点、直方图和特征值）[35]以及基于规则的方法。在建筑平面图检索中，首先需要从建筑布局中识别出基本的图形组件（如房间、门和窗）[36]。传统方法通常将文本数据与图像数据分开处理，采用手工制定的规则来识别建筑平面图中的各类元素。然而，这种依赖于人工特征设计和规则构建的方式存在诸多局限性：其一，人工设计特征与规则不仅需要专业知识，且往往难以应对不同建筑布局的复杂性和多样性，导致结果易受误差影响；其二，该方法缺乏通用性，难以适应不同的建筑风格，需针对每个具体场景进行大量定制化调整。

早在1996年，一项研究[37]就提出了一种用于自动识别与解析手绘建筑平面图的原型系统。该系统采用模式识别技术对预处理后的图像进行分析，提取建筑构件（如房间、墙体、门窗等），并将其组织为结构化格式。2011年，另一项研究[38]则利用加速稳健特征（SURF），包括边缘提取与边界检测，从建筑平面图中提取局部信息；随后，所提取的结构化数据被用于分析与检索任务。在基于内容的图像检索（CBIR）中，符号化定位技术实现了以查询图像为依据的文档检索，从而能够确定查询图像的具体位置。此外，在文献[39]中还提出了一种基于图的检索方法，即“房间连通性图”，该方法从……中提取房间连通性图。

the polygonal representations of buildings, capturing their topological relationships. The similarity between the query graph and the building model in the database was determined by evaluating the subgraph isomorphism between the query graph and the room connectivity graph. This approach, effective for smaller graphs, showed its viability for specific retrieval challenges. Another study [40] introduced a retrieval technique based on visual cues, quantifying spatial and line features using Runlength Histogram (RH). The similarity between two building floor plans was computed using the cardinal distance (χ^2 distance), with the retrieved similarity verified through subjective observer evaluation.

$$\chi^2(P,Q) = \frac{1}{2}\sum_i \frac{(P_i - Q_i)^i}{(P_i - Q_i)} \tag{10}$$

In Equation (10), P and Q represent the signatures of distinct planar graphs. With the rise in and advancement of machine learning algorithms, the inherent constraints imposed by conventional heuristics reliant on specific styles are effectively bypassed. A study published in [41] introduces a methodology grounded in statistical plane segmentation and structural pattern recognition for the analysis and interpretation of floor plans. This approach embraces a bottom–up two-step recognition procedure, wherein textual content is segregated from images through text–image segmentation during data preprocessing before the recognition phase.

The initial step of the recognition process involves identifying fundamental building region blocks, such as walls, doors, and windows, at the pixel level using statistical plane-based segmentation. Subsequently, during the transformation of the pixel image into vector space, the wall entities are isolated and amalgamated with the doors and windows, and the rooms are identified by locating enclosed areas in the solid plane to generate Figure 3, which shows the recognition process of a detailed floor plan.

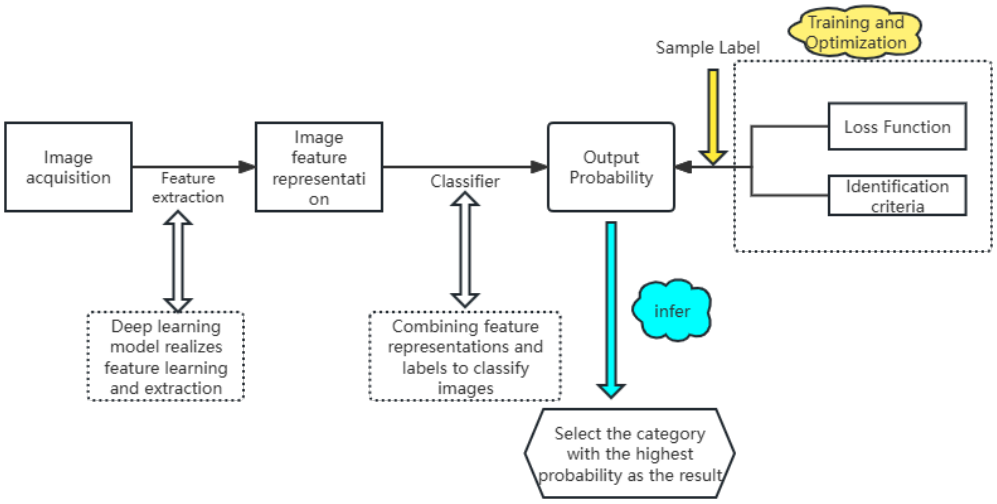


Figure 3. Building plan inspection process.

According to the China Civil Building Design Terminology Standard [42] and the Industrial Foundation Class (IFC) standard, building plans are generally categorized into two primary components: structural components and functional components. Structural components mainly include load-bearing walls, non-load-bearing walls, columns, doors, windows, railings, and stairs. In contrast, functional components, which do not directly affect the load-bearing structure but contribute to spatial partitioning and aesthetic design, include furniture elements such as tables, beds, cabinets, and other furnishings [43]. During

建筑物的多边形表示，用于刻画其拓扑关系。通过评估查询图与数据库中建筑模型的房间连通性图之间的子图同构关系，确定了两者之间的相似度。该方法在处理较小规模的图时效果显著，并展现出解决特定检索问题的可行性。另一项研究[40]提出了一种基于视觉线索的检索技术，利用游程直方图（RH）对空间特征和线状特征进行量化。两份建筑平面图之间的相似度采用卡迪纳尔距离（χ^2距离）进行计算，所得的检索结果经由主观评价予以验证。

$$\chi^2(P,Q) = \frac{1}{2}\sum_i \frac{(P_i - Q_i)^i}{(P_i - Q_i)} \tag{10}$$

在式（10）中，P 和 Q 分别表示不同平面图的特征向量。随着机器学习算法的兴起与不断进步，传统启发式方法因依赖特定模式而产生的固有约束得以有效规避。文献[41]提出了一种基于统计平面分割与结构模式识别的方法，用于对建筑平面图进行分析与解读。该方法采用自底向上的两步识别流程：在识别阶段之前，通过文本—图像分割技术在数据预处理阶段将文本内容从图像中分离出来。

识别过程的第一步是利用基于统计的平面分割方法，在像素级别识别墙体、门和窗等基本建筑单元。随后，在将像素图像转换为矢量空间的过程中，墙体实体被单独提取并与门窗合并，通过在二维平面上识别封闭区域来确定房间边界，从而生成图3，该图展示了详细平面图的识别流程。

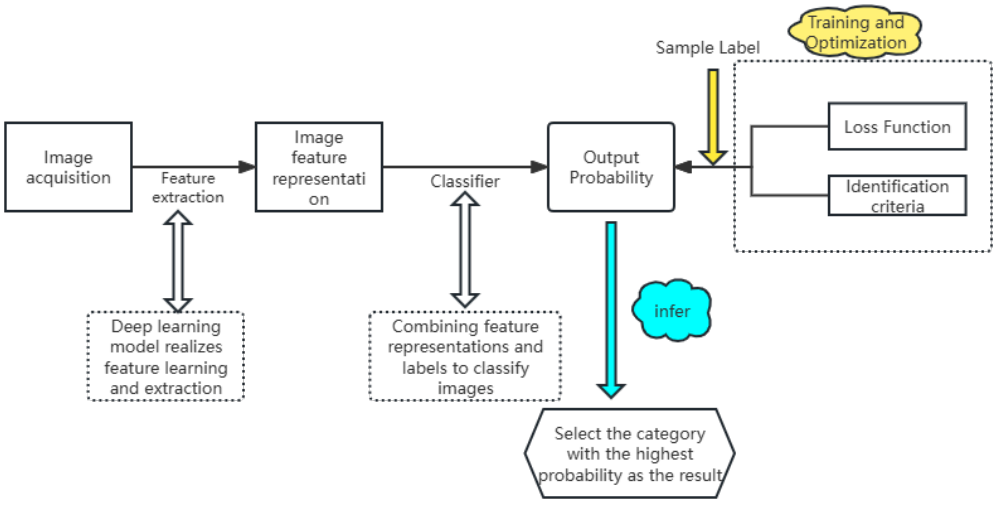


图3. 建筑平面图检查流程。

根据《中国民用建筑设计术语标准》[42]及工业基础类（IFC）标准，建筑平面通常可分为两大类：结构构件与功能构件。其中，结构构件主要包括承重墙、非承重墙、柱、门、窗、栏杆及楼梯等；而功能构件则不直接参与承重结构，但用于空间分隔与美学设计，包括桌、床、柜及其他家具等[43]。在……过程中

searches for building floor plans, these component types can be analyzed based on their characteristics and functions to improve search efficiency and accuracy.

In the field of building structural components, a study [44] introduced a lightweight and fully automated processing technique for analyzing building floor plans. Initially, the method applies regional segmentation using the mean integral projection function (IPF) to identify wall-containing regions. Subsequently, critical information, such as wall locations, is extracted using a sparse point pixel vectorization algorithm based on non-detailed data. Finally, a linear discriminative analysis algorithm, utilizing QR decomposition and generalized singular value decomposition, is employed to identify building components such as doors, windows, and wall openings. Another study [38] proposed an algorithm specifically designed to recognize and extract structural components, including walls, doors, and windows. The approach eliminates the exterior wall through successive erosion and expansion operations, extracts wall contours from connected components, applies polygon approximation to each contour to determine wall edges, and uses symbol recognition techniques such as SURF to detect doors and windows.

Structural components such as walls, doors, and windows exhibit non-distinct features and varied shapes in plan views, leading to strong abstraction and weak regularity, which can reduce direct recognition efficiency. Therefore, initial recognition of functional components in building floor plans, such as tables, beds, and chairs, becomes essential. A study [20] introduces the Faster R-CNN model, which leverages three deep learning networks to recognize, locate, and remove functional components from building floor plans. Another study [25] proposes a deep multi-neural network incorporating spatial context modules and room boundary-guided attention mechanisms to improve the recognition performance of various floor plan elements, including walls, rooms, doors, and windows.

These methodologies represent deep learning-based approaches for recognizing architectural floor plan elements. After recognizing elements, retrieving building floor plan components requires matching based on geometric features such as shape, location, and others. A study [45] proposes a sketch-based system (a.Scatch system) that extracts semantic structures from past projects. Initially, information such as walls, symbols, and text is segmented, yielding thick, medium, and thin line images through erosion and expansion operations. Thick lines delineate building boundaries, medium lines represent internal structures, and thin lines indicate architectural elements such as doors, windows, and furniture. Next, structural information is extracted and analyzed semantically using Speeded Up Robust Features (SURF). The extracted structures are then compared using graph-matching techniques to retrieve the most similar results. Another approach [46] revolves around PU learning, efficiently analyzing planar graphs to recognize various structural element styles with minimal user interaction. This method involves extracting regions of interest (RoIs) from the image, filtering RoIs similar to the query based on IoU thresholds, extracting features using Haar-based kernels on the remaining RoIs, and ultimately retrieving similar RoIs through PU learning.

4. Texture Feature Retrieval

4.1. Gabor Wavelet Transform

Texture features in an image are inherently linked to the surface structure of the depicted object. Image texture reflects local structural characteristics, primarily manifested as variations in pixel grayscale or color within a defined neighborhood. In the early 1970s, Haralick et al. introduced the co-occurrence matrix representation for characterizing texture features [47]. A co-occurrence matrix is first constructed based on pixel direction and distance, and relevant statistics are then extracted to serve as a texture representation. However, the texture properties derived from the co-occurrence matrix often fail to align

在建筑物平面图的检索过程中，可根据这些构件的特征与功能对其进行分析，以提升检索效率与准确性。

在建筑结构构件领域，一项研究[44]提出了一种轻量级且完全自动化的处理技术，用于分析建筑平面图。该方法首先利用均值积分投影函数（IPF）进行区域分割，以识别包含墙体的区域；随后，基于非细节化数据，采用稀疏点像素矢量化算法提取墙体位置等关键信息；最后，通过结合QR分解与广义奇异值分解的线性判别分析算法，实现对门、窗及墙体洞口等建筑构件的识别。另一项研究[38]则提出了一种专门用于识别和提取结构构件（包括墙体、门和窗）的算法。该方法通过连续的腐蚀与膨胀操作去除外墙，从连通域中提取墙体轮廓，对各轮廓进行多边形逼近以确定墙体边界，并借助SURF等符号识别技术检测门和窗。

墙体、门窗等结构构件在平面图中呈现出特征不明显且形状多样的特点，导致抽象程度高、规律性弱，从而降低直接识别的效率。因此，对建筑平面图中的功能构件（如桌椅、床等）进行初步识别显得尤为重要。一项研究[20]提出了基于Faster R-CNN的模型，该模型利用三个深度学习网络实现对建筑平面图中功能构件的识别、定位及去除。另一项研究[25]则提出了一种融合空间上下文模块与房间边界引导注意力机制的深度多神经网络，以提升对墙体、房间、门窗等多种平面图要素的识别性能。

这些方法代表了基于深度学习的建筑平面图元素识别技术。在完成元素识别后，建筑平面图构件的检索需要依据形状、位置等几何特征进行匹配。一项研究[45]提出了一种基于草图的系统（a.Scatch系统），该系统可从既有项目中提取语义结构。首先对墙体、符号及文本等信息进行分割，并通过腐蚀与膨胀操作生成粗线、中线和细线图像：其中，粗线用于勾勒建筑边界，中线表示内部结构，细线则标识门、窗及家具等建筑构件。随后，利用加速稳健特征（SURF）对结构信息进行语义提取与分析，并借助图匹配技术对所提取的结构进行比对，以检索出最相似的结果。另一种方法[46]则基于部分标记学习（PU学习），通过高效地分析平面图结构，仅需少量用户交互即可识别多种结构元素的样式。该方法首先从图像中提取感兴趣区域（RoI），根据交并比（IoU）阈值筛选与查询相似的RoI；然后在剩余的RoI上使用基于Haar核的特征提取算法提取特征，并最终通过部分标记学习检索出相似的RoI。

4. 纹理特征检索

4.1. Gabor Wavelet Transform

图像中的纹理特征本质上与所描绘对象的表面结构密切相关。图像纹理反映了局部的结构特性，主要表现为在给定邻域内像素灰度或颜色的变化。20世纪70年代初，Haralick等人提出了用于表征纹理特征的共生矩阵方法[47]。首先根据像素的方向和距离构建共生矩阵，然后提取相应的统计量作为纹理的表征。然而，由共生矩阵得到的纹理特征往往难以与实际需求相匹配。

with visual similarity. As a result, there is a need to explore feature extraction methods that better align with human visual perception.

In the early 1990s, following the introduction and theoretical foundation of the wavelet transform, numerous researchers began investigating its application in texture representation. Empirical evidence has shown the effectiveness of these studies. Manjunath and Ma [48] showed that using the Gabor wavelet transform for feature extraction in content-based image retrieval is more effective than employing the pyramid wavelet transform (PWT) or the tree wavelet transform (TWT). The Gabor filter consists of a collection of wavelets, each capturing energy at a specific frequency and orientation. According to this framework, the signal is expanded to provide a local frequency description, effectively capturing its local characteristics and energy. Texture features can then be extracted from this collection of energy distributions. The variability in scale (frequency) and direction of the Gabor filter makes it particularly useful for texture analysis.

For an image defined with size $P \times Q$, its high-frequency Gabor wavelet transform is represented as

$$G_{mm}(x,y) = \sum_{s}\sum_{t} I(x-s, y-t)\Psi^{*}_{mn}(s,t) \tag{11}$$

where s and t are the variables of the wavelet filter size, and Ψ^{*}_{mn} is the complex transform of Ψ_{mn}. Ψ_{mn} is derived from Equation (12), which is the wavelet function, generating a set of similar functions, defined as

$$\Psi(x,y) = \frac{1}{2\pi\sigma_x\sigma_y}\exp\left(-\frac{1}{2}\left(\frac{x^2}{\sigma_x^2}+\frac{y^2}{\sigma_y^2}\right)\right)\cdot\exp(j2\pi Wx) \tag{12}$$

Among them, W is referred to as the modulation frequency. It is derived from the Gabor wavelet equation (Equation (13)):

$$\Psi_{mn}(x,y) = a^{-m}\Psi(\tilde{x},\tilde{y}) \tag{13}$$

Here, m and n represent the standards and directions of the small waves, where $m = 0, 1, \ldots, M-1$ and $n = 0, 1, \ldots, N-1$:

$$\tilde{x} = a^{-m}(x\cos\theta + y\sin\theta) \tag{14}$$

$$\tilde{y} = a^{-m}(x\cos\theta + y\sin\theta) \tag{15}$$

In the equation, $a > 1, \theta = \mathrm{r}\pi/N$. The variables in Equations (12)–(15) are defined as follows:

$$\begin{aligned} a &= (U_h/U_l)^{\frac{1}{M-1}} \\ W_{m,n} &= a^m U_I \\ \sigma_{y,m,n} &= \frac{1}{2\pi\tan\left(\frac{\pi}{2N}\right)\sqrt{\frac{U_h}{2\ln 2}-\left(\frac{1}{2\pi\sigma_n}\right)^2}} \end{aligned} \tag{16}$$

4.2. Texture Spectrum

After performing multi-scale and multi-directional filtering on an image, a multidimensional array is obtained:

$$\begin{aligned} &E(x,y) = \sum_{x}\sum_{y}|G_{mn}(x,y)|, m = 0, 1, \cdots \\ &M-1, n = 0, 1, \cdots, N-1 \end{aligned} \tag{17}$$

具有视觉相似性。因此，有必要探索更符合人类视觉感知的特征提取方法。

20世纪90年代初，在小波变换被引入并奠定其理论基础之后，众多研究者开始探索其在纹理表示中的应用。实证研究表明，这些研究具有较高的有效性。Manjunath和Ma[48]指出，在基于内容的图像检索中，采用Gabor小波变换进行特征提取的效果优于使用金字塔小波变换（PWT）或树形小波变换（TWT）。Gabor滤波器由一组小波构成，每个小波分别捕获特定频率和方向上的能量。根据这一框架，信号被展开以提供局部频率描述，从而有效捕捉其局部特征与能量分布。随后，可从这一系列能量分布中提取纹理特征。Gabor滤波器在尺度（频率）和方向上的可变性使其在纹理分析中尤为适用。

对于一幅尺寸为 $P \times Q$ 的图像，其高频Gabor小波变换可表示为

$$G_{mm}(x,y) = \sum_{s}\sum_{t} I(x-s, y-t)\Psi^{*}_{mn}(s,t) \tag{11}$$

其中，s 和 t 为小波滤波器的尺度参数，Ψ^{*}_{mn} 为 Ψ_{mn} 的复数变换。Ψ_{mn} 由式（12）导出，该式为小波函数，并由此生成一组相似函数，定义如下：

$$\Psi(x,y) = \frac{1}{2\pi\sigma_x\sigma_y}\exp\left(-\frac{1}{2}\left(\frac{x^2}{\sigma_x^2}+\frac{y^2}{\sigma_y^2}\right)\right)\cdot\exp(j2\pi Wx) \tag{12}$$

其中，W被称为调制频率。其源自Gabor小波方程（式（13））：

$$\Psi_{mn}(x,y) = a^{-m}\Psi(\tilde{x},\tilde{y}) \tag{13}$$

此处，m 和 n 分别表示小波的标准和方向，其中 $m = \in \{0, 1, \cdots\}$，$M - \in \{0, 1\}$，且 $n = \in \{0, 1, \cdots\}$，$N - \in \{0, 1\}$。

$$\tilde{x} = a^{-m}(x\cos\theta + y\sin\theta) \tag{14}$$

$$\tilde{y} = a^{-m}(x\cos\theta + y\sin\theta) \tag{15}$$

在该方程中，$a > = 1$，$\theta = = \pi/N$。式（12）至式（15）中的各变量定义如下：

$$\begin{aligned} a &= (U_h/U_l)^{\frac{1}{M-1}} \\ W_{m,n} &= a^m U_I \\ \sigma_{y,m,n} &= \frac{1}{2\pi\tan\left(\frac{\pi}{2N}\right)\sqrt{\frac{U_h}{2\ln 2}-\left(\frac{1}{2\pi\sigma_n}\right)^2}} \end{aligned} \tag{16}$$

4.2. Texture Spectrum

在对图像进行多尺度和多方向滤波处理后，可得到一个多维数组：

$$\begin{aligned} &E(x,y) = \sum_{x}\sum_{y}|G_{mn}(x,y)|, m = 0, 1, \cdots \\ &M-1, n = 0, 1, \cdots, N-1 \end{aligned} \tag{17}$$

These dimensions represent the energy of different scales and directions in the image. The main purpose of texture-based image retrieval is to find images or regions in images with similar textures. Assuming that an image or a region has the same texture, the mean value μ_{mn} and standard deviation σ_{mn} of the transform coefficients can be used to represent the texture characteristics of the region:

$$\begin{aligned} \mu_{mn} &= \frac{E(x,y)}{P \times Q} \\ \sigma_{mn} &= \frac{\sqrt{\sum_x \sum_y (\mid G_{mn}(x,y) \vdash \mu_{mn})^2}}{P \times O} \end{aligned} \tag{18}$$

He DC et al. proposed the concept of texture spectrum in the early 1990s [49]. Compared with previous methods, texture spectrum has a clearer concept and requires less computation, and has received more and more attention in recent years. The following will describe in detail the basic method of texture spectrum model and some improved texture spectrum models. For a 3×3 neighborhood in an image, as shown in Table 1, the nine pixels in the neighborhood are recorded as $V = V_0, V_1, V_2, V_3, V_4, V_5, V_6, V_7, V_8$, where V_0 represents the pixel value at the center of the neighborhood. At the same time, a texture unit T_U is defined, and T_U contains eight pixels. $T_U = E_1, E_2, E_3, E_4, E_5, E_6, E_7, E_8$, where the value of E_i is

$$E_i \begin{cases} 0 & V_i \leq (V_0 - C) \\ 1 & (V_0 + C) \geq V_i \geq (V_0 - C) \\ 2 & V_i \leq (V_0 + C) \end{cases} \tag{19}$$

where $i = 1, 2, \cdots, 8$, C; C represents a positive constant; and E_i corresponds to pixel i. For each element in TU, there are three possible values, so for 8 units, there are $3^8 = 6561$ possible values. The texture unit can be written as follows:

$$\begin{aligned} N_{TU} &= \sum_{i=1}^{8} E_i \times 3^{i-1} \\ N_{TU} &\in \{0, 1, 2, \cdots, 6560\} \end{aligned} \tag{20}$$

An image texture unit characterizes the textural features of a given pixel, specifically the relative grayscale relationship between the central pixel and its neighboring pixels. The occurrence frequency of all texture units in the image is quantified, and this frequency function delineates the image's texture information. This function, called the texture spectrum, encapsulates the distribution of all texture units. A texture spectrum histogram can be created, with the horizontal axis representing the number of texture units (NTUs) and the vertical axis indicating their occurrence frequency. Typically, an image consists of two components: texture primitives and random noise or background. The greater the proportion of texture components relative to the background, the more easily the texture features can be perceived by the human eye. With respect to the texture spectrum, a higher percentage of texture components in an image results in a distinct peak distribution. Additionally, different textures are characterized by unique texture units and corresponding texture spectra. Thus, the texture information of an image is effectively represented by the texture spectrum.

Table 1. Texture unit.

V_1	V_2	V_3
$V_4 V_6$	$V_0 V_7$	$V_5 V_8$

这些尺度特征分别表征了图像中不同尺度和方向的能量。基于纹理的图像检索的主要目标是查找具有相似纹理的图像或图像中的区域。假设某图像或某区域具有相同的纹理，则可通过其变换系数的均值μ_{mn}和标准差σ_{mn}来表征该区域的纹理特征：

$$\begin{aligned} \mu_{mn} &= \frac{E(x,y)}{P \times Q} \\ \sigma_{mn} &= \frac{\sqrt{\sum_x \sum_y (\mid G_{mn}(x,y) \vdash \mu_{mn})^2}}{P \times O} \end{aligned} \tag{18}$$

He DC 等人在20世纪90年代初提出了纹理谱的概念[49]。与以往的方法相比，纹理谱的概念更加清晰，计算量也更小，近年来受到了越来越多的关注。下文将详细阐述纹理谱模型的基本方法以及一些改进的纹理谱模型。对于图像中的一个3×3邻域，如表1所示，该邻域内的九个像素分别记为v5、v6、v7、v8、v9、v10、v11、v12、v13，其中v14表示邻域中心的像素值。同时，定义一个纹理单元v15，该单元包含八个像素v16、v17、v18、v19、v20、v21、v22、v23、v24，其中v25的值为

$$E_i \begin{cases} 0 & V_i \leq (V_0 - C) \\ 1 & (V_0 + C) \geq V_i \geq (V_0 - C) \\ 2 & V_i \leq (V_0 + C) \end{cases} \tag{19}$$

其中i =取值为1、2、⋯、8、C；C表示一个正的常数；E_i对应于像素i。对于TU中的每个元素，存在三种可能的取值，因此对于8个单元，共有3^8 =，即6561种可能的取值组合。纹理单元可表示如下：

$$\begin{aligned} N_{TU} &= \sum_{i=1}^{8} E_i \times 3^{i-1} \\ N_{TU} &\in \{0, 1, 2, \cdots, 6560\} \end{aligned} \tag{20}$$

图像纹理单元用于表征给定像素的纹理特征，具体而言，即中心像素与其邻域像素之间的相对灰度关系。对图像中所有纹理单元的出现频率进行量化，该频率函数即可刻画图像的纹理信息。这一函数被称为纹理谱，它完整地描述了所有纹理单元的分布情况。基于纹理谱可构建纹理谱直方图，其中横轴表示纹理单元的数量（NTU），纵轴则表示其出现频率。通常，一幅图像由两类成分构成：纹理基元与随机噪声或背景。当纹理成分相对于背景所占比例越大时，其纹理特征就越容易被人类视觉所感知。就纹理谱而言，图像中纹理成分占比越高，其峰态分布越为显著。此外，不同纹理分别对应独特的纹理单元及相应的纹理谱。因此，图像的纹理信息可通过纹理谱得到有效表征。

表1. 质地单位。

V_1	V_2	V_3
V_4V_6	V_0V_7	V_5V_8

5. Spatial Feature Retrieval

5.1. Topological and Geometric Feature

5.1.1. Topological Feature Retrieval

Conventional methods for locating room layouts primarily focus on matching room functions and adjacency, often overlooking the internal shapes of rooms. Inspired by shape grammar principles, Lee et al. introduce a novel approach that transforms building floor plans into a tree-like structure to address retrieval challenges [50]. This method integrates internal layout considerations into similarity metrics and represents room layout structures through a hierarchical tree framework. Floor plan retrieval is then executed using tree editing distance as a metric for query similarity. Tree structure matching is computationally efficient, preserving high-level feature information through parent–child relationships. This retrieval method, based on matching the shapes of structural hierarchies, eliminates the need to account for variations in room functions, resulting in a remarkable experimental efficacy of 53%. The tree editing distance [27] serves as the key similarity metric between the query and the candidate parse tree. To compute the tree edit distance effectively, costs must be allocated to editing operations (relabeling, adding, deleting), requiring an ordered arrangement of the tree to achieve polynomial complexity [31,51]. Initially, tree nodes are sorted based on the number of child nodes and, in cases of equal child nodes, by room size. The matching metric C is computed as follows:

$$C(A, B) = ||area(A)v_A - area(B)v_B||_1 \tag{21}$$

In Equation (21), $area(A)$ denotes the area covered by the parameterized shape at the node, while $\|*\|_1$ represents the mean absolute difference.

5.1.2. Geometric Feature Retrieval

The shape-based retrieval method offers several advantages over feature extraction methods. It effectively handles transformations like translation, rotation, and scaling, improving retrieval performance for complex building plan shapes. However, this method has limitations, including mismatching building plans with similar shapes but different scales and faces difficulty in extracting shape features from non-geometric building plans. To address these issues, Rasika et al. proposed a multi-directional building plan retrieval approach using FR-CNN and geometric features [52]. This method accounts for rotation during scanning and is invariant to scale changes. The process involves detecting door endpoints on walls. After removing the door from the image, a blank area remains on the outer wall of the I_F. Next, the endpoints of the door are connected with a straight line to ensure continuity. Finally, the shape of I_F is obtained by tracing the outer wall. Geometric features such as area, corners, axes, distances, slopes, and angles are computed. These geometric features are interdependent, and polygonal approximation is used to represent the extracted profile.

The matching phase includes matching based on appearance features (F_1) and matching based on internal object features (F_1). For the matching based on profile features, the query image (F^q) and the dataset image (F^i) are matched using Equation (12), and the average matching cost is calculated using Equation (13), where DS denotes the size of the dataset image, $j \in \{1, 2\}$, F_j denotes the jth feature, and n_j denotes its corresponding count. Let F_{jk} denote the kth component of the jth feature. Here, the matching cost based on the shape features φ when $F_j = F_1$ is $n_j = n_1$. If $j = 1$, then F_{jk} means that the feature is profile-based, and k denotes the index of the feature component.

$$\psi\left(F_{jk}\right)^i = \left[\left[F^i_{jk} - F^q_{jk}\right]_{k=1}^{n_j}\right]_{i=1}^{DS} \tag{22}$$

5. 空间特征检索

5.1. Topological and Geometric Feature

5.1.1. 拓扑特征检索

用于定位房间布局的传统方法主要侧重于房间功能与邻接关系的匹配，往往忽视了房间的内部形态。受形状文法规则的启发，Lee等人提出了一种新方法，将建筑平面图转换为树状结构，以应对检索难题[50]。该方法将内部布局因素纳入相似性度量，并通过层次化树框架来表征房间布局结构。随后，以树编辑距离作为查询相似性的度量指标进行平面图检索。树结构匹配具有较高的计算效率，并可通过父子关系保留高层次特征信息。这种基于结构层次形态匹配的检索方法无需考虑房间功能的差异，实验结果表明其检索效率高达53%。其中，树编辑距离[27]是查询树与候选解析树之间的重要相似性度量。为有效计算树编辑距离，需为各类编辑操作（重新标记、插入、删除）分配相应的代价，并对树节点进行有序排列，从而将计算复杂度控制在多项式级别[31,51]。初始阶段，树节点按子节点数量进行排序；当子节点数量相同时，则按房间面积大小排序。匹配度量C的计算公式如下：

$$C(A,B) = ||area(A)v_A - area(B)v_B||_1 \tag{21}$$

在式（21）中，$area(A)$ 表示参数化形状在该节点处所覆盖的面积，而 $\|*\|_1$ 则表示平均绝对差。

5.1.2 几何特征检索

基于形状的检索方法相较于特征提取方法具有多项优势。它能够有效处理平移、旋转和缩放等变换，从而提升复杂建筑平面形状的检索性能。然而，该方法也存在局限性，例如难以匹配形状相似但尺度不同的建筑平面，且在从非几何型建筑平面中提取形状特征时面临困难。为解决这些问题，Rasika 等人提出了一种基于 FR-CNN 和几何特征的多方向建筑平面检索方法[52]。该方法在扫描过程中考虑了旋转因素，并对尺度变化具有不变性。具体流程包括：首先检测墙面上的门端点；随后将门从图像中去除，在I_F的外墙上留下一个空白区域；接着通过直线连接门的两个端点以保证连续性；最后，沿外墙轮廓提取I_F的形状。在此基础上，计算面积、角点、主轴、距离、斜率及角度等几何特征。这些几何特征之间相互关联，并采用多边形逼近法对所提取的轮廓进行表示。

匹配阶段包括基于外观特征的匹配（F_1）和基于物体内部特征的匹配（F_1）。对于基于轮廓特征的匹配，查询图像（F^q）与数据集图像（F^i）采用式（12）进行匹配，并通过式（13）计算平均匹配代价，其中 DS 表示数据集图像的尺寸，$j \in \{1, 2\}$ 表示第 j 个特征，F_j 表示该特征的对应计数，n_j 表示其对应的权重。令 F_{jk} 表示第 j 个特征的第 k 个分量。此处，基于形状特征的匹配代价 φ 在 $F_j = F_1$ 为 $n_j = n_1$ 时计算。若 $j =$ 等于 1，则 F_{jk} 表示该特征为轮廓特征，k 表示该特征分量的索引。

$$\psi\left(F_{jk}\right)^i = \left[\left[F_{jk}^i - F_{jk}^q\right]_{k=1}^{n_j}\right]_{i=1}^{DS} \tag{22}$$

$$Avg(F_j)^i = \left[\frac{1}{n_j}\sum_{k=1}^{n_j}\psi\left(F_{jk}\right)^i\right]_{i=1}^{DS} \quad (23)$$

The average feature matching cost (Avg) is calculated using Equation (22) and sorted in ascending order using Equation (23) to prioritize maximum similarity.

$$\varphi(j)^i = \left[sort_{asc}\left(Avg(F_j)^i\right)\right]_{i=1}^{DS} \quad (24)$$

For internal object-based feature matching, the sorted results of external shape feature matching prior to the last image associated with the query are considered when calculating the matching cost, rather than the entire dataset. This is the case, and when $F_j = F_2$, $n_j = n_2$ after calculating the matching cost using Equation (22). Further, the average matching cost for internal object features is calculated using Equation (23). Finally, the retrieval results are sorted in ascending order using Equation (24) to obtain a final ranked list of the query images, which includes the images before the last relevant image of the query image.

5.2. Multidimensional and Shape Feature Retrieval

5.2.1. Multidimensional Feature Retrieval

To expand content-based image retrieval technology to accommodate a large-scale image library, the adoption of effective multidimensional indexing techniques is essential. The challenges associated with this expansion can be categorized into two main aspects:

1. High dimensionality: Typically, the dimension of the image feature vector is on the order of 10^2.
2. Non-Euclidean similarity measurement: The Euclidean distance metric often fails to adequately represent all human perceptions of visual content; consequently, alternative similarity measurement methods, such as histogram intersection, cosine similarity, and correlation, must be employed.

To address these challenges, a viable approach involves initially applying dimensionality reduction techniques to minimize the feature vector's dimensionality, followed by the utilization of suitable multidimensional indexing methods.

Three primary research domains have significantly advanced the development of multidimensional indexing technology: computational geometry, database management systems, and pattern recognition. Prominent multidimensional indexing technologies currently in use include the Bucketing grouping algorithm, k-d trees, priority k-d trees, quadtrees, K-D-B trees, HB trees, R-trees, and their variants, such as the R+ tree and R* tree. Additionally, clustering and neural network techniques, which are widely employed in pattern recognition, also offer viable indexing solutions. The origins of multidimensional indexing technology can be traced to the mid-1970s, when foundational techniques such as the Cell algorithm, quadtrees, and k-d trees were first introduced, albeit with limited initial effectiveness. The growing demand for spatial indexing in Geographic Information Systems (GISs) and Computer-Aided Design (CAD) systems culminated in Guttman's proposal of the R-tree index structure in 1984 [53]. This innovation spurred the development of several R-tree variants, including Sellis et al.'s R+ tree [54] and Greene's R-tree variant [55]. In 1990, Beckman and Kriegel introduced the R* tree, which is widely recognized as the most efficient dynamic R-tree variant [56]. Nevertheless, even the R* tree faces challenges in effectively managing dimensionalities exceeding 20.

A comprehensive review and comparison of various indexing algorithms are provided in references [57]. Among these, White and Jain focus on the development of general and domain-specific indexing algorithms. Inspired by k-d trees and R-trees, they proposed VAM k-d trees and VAMSplit R-trees. Experimental results demonstrate that VAMSplit R-trees

$$Avg(F_j)^i = \left[\frac{1}{n_j}\sum_{k=1}^{n_j}\psi\left(F_{jk}\right)^i\right]_{i=1}^{DS} \quad (23)$$

平均特征匹配代价（Avg）通过式（22）计算，并按式（23）进行升序排序，以优先考虑最大相似度。

$$\varphi(j)^i = \left[sort_{asc}\left(Avg(F_j)^i\right)\right]_{i=1}^{DS} \quad (24)$$

对于基于对象的内部特征匹配，在计算匹配代价时，仅考虑与查询图像关联的倒数第二张图像之前的外部形状特征匹配排序结果，而非整个数据集。具体而言，当使用式（22）计算$F_j = F_2$和$n_j = n_2$的匹配代价时即为此情形。此外，基于式（23）计算对象内部特征的平均匹配代价。最后，通过式（24）对检索结果按升序进行排序，得到查询图像的最终排序列表，该列表包含查询图像相关性最高的图像之前的所有图像。

5.2. Multidimensional and Shape Feature Retrieval

5.2.1 多维特征检索

为使基于内容的图像检索技术能够适应大规模图像库，采用有效的多维索引技术至关重要。与此扩展相关的挑战可归纳为两个主要方面：

1. 高维性：通常，图像特征向量的维度数量级为10 。2. 非欧氏相似性度量：欧氏距离度量往往无法充分反映人类对视觉内容的所有感知；因此，必须采用诸如直方图交集、余弦相似度和相关系数等替代的相似性度量方法。

为应对这些挑战，一种可行的方案是首先采用降维技术以降低特征向量的维度，随后再利用合适的多维索引方法。

三个主要的研究领域显著推动了多维索引技术的发展：计算几何、数据库管理系统以及模式识别。目前广泛使用的多维索引技术包括分桶分组算法、k-d树、优先级k-d树、四叉树、K-D-B树、HB树、R树及其变体，如R+树和R*树。此外，模式识别中广泛应用的聚类技术和神经网络方法也为索引提供了可行的解决方案。多维索引技术的起源可追溯至20世纪70年代中期，当时诸如单元算法、四叉树和k-d树等基础技术被首次提出，但其初期效果较为有限。随着地理信息系统（GIS）和计算机辅助设计（CAD）系统对空间索引需求的日益增长，Guttman于1984年提出了R树索引结构[53]，这一创新促进了多种R树变体的开发，其中包括Sellis等人提出的R+树[54]以及Greene的R树变体[55]。1990年，Beckman和Kriegel提出了R*树，该结构被公认为最高效的动态R树变体[56]。然而，即便是R*树，在处理超过20维的数据时仍面临挑战。

对各类索引算法的全面综述与比较见文献[57]。其中，White和Jain着重研究通用及领域特定索引算法的开发。受k-d树和R树的启发，他们提出了VAM k-d树和VAMSplit R树。实验结果表明，VAMSplit R树

exhibit superior algorithmic efficiency, albeit at the cost of sacrificing the dynamic characteristics inherent to R-trees. In contrast, Ng and Sedighian [58] introduced a three-stage retrieval technique for image retrieval, comprising dimensionality reduction, evaluation and selection of existing indexing algorithms, and optimization of the chosen indexing algorithm. Since nearly all tree-structured indexing techniques are primarily designed for traditional database queries (e.g., point queries and range queries) rather than image retrieval, there is an urgent need to explore novel indexing structures tailored to the specific requirements of image retrieval. Tagare addressed this challenge in [59] by proposing a method for tree structure adjustment, which improves tree efficiency by eliminating nodes that hinder the effectiveness of similarity queries.

5.2.2. Fourier Shape Descriptor

Shape features can generally be represented in two ways: as contour features or regional features. Contour features focus solely on the outer boundary of an object, while regional features encompass the entire area of the shape. The most common methods for these two types of shape features are Fourier descriptors and shape-independent moments.

The fundamental concept of Fourier shape descriptors is to utilize the Fourier transform of the object's boundary for shape representation. Consider the contour of a two-dimensional object composed of a series of pixels with coordinates (x_s, y_s), where $0 \leq s \leq N-1$, and N represents the total number of pixels on the contour. From the coordinates of these boundary points, three shape representations can be derived: the curvature function, centroid distance, and complex coordinate function. The curvature of a point on a profile is defined as the rate of change in the profile tangent angle relative to the arc length. The curvature function $K(s)$ can be expressed as $K(s) = \frac{d}{ds}\theta(s)$, where $\theta(s)$ is the tangent angle of the contour line, defined as

$$\left.\begin{array}{l} \theta(s) = \tan^{-1}\left(\frac{y'_s}{x'_s}\right) \\ y'_s = \frac{dy_s}{ds} \\ x'_s = \frac{dx_s}{ds} \end{array}\right\} \tag{25}$$

The centroid distance is defined as the distance from the object's boundary point to the object's center (x_c, y_c), as follows:

$$R(s) = \sqrt{(x_s - x_c)^2 + (y_s - y_c)^2} \tag{26}$$

The complex coordinate function is the pixel coordinate expressed by complex numbers:

$$Z(s) = (x_s - x_c) + j(y_s - y_c) \tag{27}$$

The Fourier transform of the complex coordinate function generates a series of complex coefficients. These coefficients characterize the object's shape in the frequency domain, with low-frequency components indicating macroscopic properties and high-frequency components detailing fine structural characteristics. A shape descriptor can be derived from these transformation parameters. To maintain rotation independence, only the magnitude information of the parameters is retained, while the phase information is omitted. Scaling independence is achieved by normalizing the magnitude of the parameter to the magnitude of the DC component (or the first non-zero parameter).

For the curvature function and the centroid distance function, we focus solely on the positive frequency coordinate axis because the Fourier transform of these functions

尽管以牺牲R树固有的动态特性为代价，这些方法仍表现出卓越的算法效率。相比之下，Ng和Sedighian[58]提出了一种用于图像检索的三阶段检索技术，包括降维、现有索引算法的评估与选择，以及所选索引算法的优化。由于几乎所有的树结构索引技术主要针对传统数据库查询（如点查询和范围查询）而设计，而非专门用于图像检索，因此迫切需要探索能够满足图像检索特定需求的新型索引结构。Tagare在[59]中通过提出一种树结构调整方法来应对这一挑战，该方法通过删除那些妨碍相似性查询效率的节点，从而提升树结构的检索效率。

5.2.2 傅里叶形状描述子

形状特征通常可以通过两种方式表示：轮廓特征和区域特征。轮廓特征仅关注物体的外边界，而区域特征则涵盖整个形状的区域。对于这两种类型的形状特征，最常用的方法分别是傅里叶描述子和形状不变矩。

傅里叶形状描述子的基本思想是利用物体边界轮廓的傅里叶变换来进行形状表示。考虑一个由一系列像素点组成的二维物体的轮廓，其像素点的坐标为(x_s, y_s)，其中$0 \leqslant \leq s \leq N - \leqslant 1$，$N$表示轮廓上像素点的总数。从这些边界点的坐标出发，可以导出三种形状表示：曲率函数、质心距离和复坐标函数。轮廓上某一点的曲率定义为轮廓切线角相对于弧长的变化率。曲率函数$K(s)$可表示为$K(s) = \frac{d}{ds}\theta(s)$，其中$\theta(s)$为轮廓线的切线角，其定义为

$$\left.\begin{array}{l} \theta(s) = \tan^{-1}\left(\frac{y'_s}{x'_s}\right) \\ y'_s = \frac{dy_s}{ds} \\ x'_s = \frac{dx_s}{ds} \end{array}\right\} \tag{25}$$

质心距离被定义为物体边界点到物体中心的距离（x_c，y_c），具体如下：

$$R(s) = \sqrt{(x_s - x_c)^2 + (y_s - y_c)^2} \tag{26}$$

复数坐标函数是用复数表示的像素坐标：

$$Z(s) = (x_s - x_c) + j(y_s - y_c) \tag{27}$$

复坐标函数的傅里叶变换会产生一系列复系数。这些系数在频域中表征了物体的形状，其中低频分量反映宏观特征，高频分量则刻画精细的结构细节。可由这些变换参数构造形状描述子。为保持旋转不变性，仅保留参数的幅值信息，而舍弃相位信息。通过将参数幅值归一化至DC分量（或第一个非零参数）的幅值，实现尺度不变性。

对于曲率函数和质心距离函数，我们仅关注正频率坐标轴，因为这些函数的傅里叶变换

is symmetric, specifically, $|F_{-i}| = |F_i|$. The shape descriptor derived from the curvature function is formulated as follows:

$$f_K = [|F_1|, |F_2|, \ldots, |F_{M/2}|] \tag{28}$$

where F_i represents the i-th component of the Fourier transform parameter. Similarly, the shape descriptor derived from the centroid distance function is

$$f_R = \left[\frac{|F_1|}{|F_0|}, \frac{|F_2|}{|F_0|}, \ldots, \frac{|F_{M/2}|}{|F_0|}\right] \tag{29}$$

In the complex coordinate function, both positive- and negative-frequency components are utilized. The DC component is omitted as it pertains to the position of the shape. Consequently, the first non-zero frequency component is employed to normalize the remaining transformation parameters. The shape descriptor derived from the complex coordinate function is formulated as follows:

$$f_Z = \left[\frac{\left|F_{-(M/2-1)}\right|}{|F_1|}, \ldots, \frac{|F_{-1}|}{|F_1|}, \frac{|F_2|}{|F_1|}, \ldots, \frac{|F_{M/2}|}{|F_1|}\right] \tag{30}$$

To ensure uniformity in the shape features of all objects in the database, the number of boundary points must be standardized to M before implementing the Fourier transform. For instance, M can be set as $2^n = 64$ to leverage the fast Fourier transform, thereby enhancing the algorithm's efficiency.

5.2.3. Shape-Independent Moments

Moment invariants are a region-based technique for representing the shapes of objects. Assuming R is an object represented by a binary image, the $p+q$-th-order central moment of its shape is

$$\mu_{p,q} = \sum_{(x,y)\in R} (x - x_c)^p (y - y_c)^q \tag{31}$$

where (x_c, y_c) is the center of the object. To achieve scale independence, the central moment can be normalized:

$$\eta_{p,q} = \frac{\mu_{p,q}}{\mu_{0,0}^{\gamma}}, \quad \gamma = \frac{p+q+2}{2} \tag{32}$$

Based on these moments, Hu [60] proposed a series of seven moments that are independent of translation, rotation, and scaling:

$$\begin{aligned}
\phi_1 =& \mu_{2,0} + \mu_{0,2} \\
\phi_2 =& (\mu_{2,0} - \mu_{0,2})^2 + 4\mu_{1,1}^2 \\
\phi_3 =& (\mu_{3,0} - 3\mu_{1,2})^2 + (\mu_{0,3} - 3\mu_{2,1})^2 \\
\phi_4 =& (\mu_{3,0} + \mu_{1,2})^2 + (\mu_{0,3} + \mu_{2,1})^2 \\
\phi_5 =& (\mu_{3,0} - 3\mu_{1,2})(\mu_{3,0} + \mu_{1,2})\left[(\mu_{3,0} + \mu_{1,2})^2 - 3(\mu_{0,3} + \mu_{2,1})^2\right] \\
&+ (\mu_{0,3} - 3\mu_{2,1})(\mu_{0,3} + \mu_{2,1})\left[(\mu_{0,3} + \mu_{2,1})^2 - 3(\mu_{3,0} + \mu_{1,2})^2\right] \\
\phi_6 =& (\mu_{2,0} - \mu_{0,2})\left[(\mu_{3,0} + \mu_{1,2})^2 - (\mu_{0,3} + \mu_{2,1})^2\right] + 4\mu_{1,1}(\mu_{3,0} + \mu_{1,2})(\mu_{0,3} + \mu_{2,1}) \\
\phi_7 =& (3\mu_{2,1} - \mu_{0,3})(\mu_{3,0} + \mu_{1,2})\left[(\mu_{3,0} + \mu_{1,2})^2 - 3(\mu_{0,3} + \mu_{2,1})^2\right] \\
&+ (\mu_{3,0} - 3\mu_{2,1})(\mu_{0,3} + \mu_{2,1})\left[(\mu_{0,3} + \mu_{2,1})^2 - 3(\mu_{3,0} + \mu_{1,2})^2\right]
\end{aligned} \tag{33}$$

是关于$|F_{-i}| = |F_i|$对称的。由曲率函数导出的形状描述子可表示为如下形式：

$$f_K = [|F_1|, |F_2|, \ldots, |F_{M/2}|] \tag{28}$$

其中，F_i 表示傅里叶变换参数的第 i 个分量。类似地，由质心距离函数导出的形状描述子为

$$f_R = \left[\frac{|F_1|}{|F_0|}, \frac{|F_2|}{|F_0|}, \ldots, \frac{|F_{M/2}|}{|F_0|}\right] \tag{29}$$

在复坐标函数中，同时利用了正频率和负频率分量。由于DC分量与形状的位置相关，故将其予以省略。因此，采用第一个非零频率分量对剩余的变换参数进行归一化处理。由复坐标函数导出的形状描述子可表示为：

$$f_Z = \left[\frac{\left|F_{-(M/2-1)}\right|}{|F_1|}, \ldots, \frac{|F_{-1}|}{|F_1|}, \frac{|F_2|}{|F_1|}, \ldots, \frac{|F_{M/2}|}{|F_1|}\right] \tag{30}$$

为确保数据库中所有目标的形状特征具有一致性，在实施傅里叶变换之前，需将边界点的数量标准化至M。例如，可将M设置为$2^n = 64$，以利用快速傅里叶变换，从而提升算法效率。

5.2.3. 与形状无关的矩

矩不变量是一种基于区域的物体形状表示方法。假设R为由二值图像表示的物体，则其形状的$p+q$-阶中心矩为

$$\mu_{p,q} = \sum_{(x,y)\in R} (x - x_c)^p (y - y_c)^q \tag{31}$$

其中，(x_c, y_c) 为该物体的质心。为了实现尺度不变性，可对中心矩进行归一化处理：

$$\eta_{p,q} = \frac{\mu_{p,q}}{\mu_{0,0}^{\gamma}}, \quad \gamma = \frac{p+q+2}{2} \tag{32}$$

基于这些矩，Hu [60] 提出了一组与平移、旋转和缩放无关的七个矩：

$$\begin{aligned}
\phi_1 =& \mu_{2,0} + \mu_{0,2} \\
\phi_2 =& (\mu_{2,0} - \mu_{0,2})^2 + 4\mu_{1,1}^2 \\
\phi_3 =& (\mu_{3,0} - 3\mu_{1,2})^2 + (\mu_{0,3} - 3\mu_{2,1})^2 \\
\phi_4 =& (\mu_{3,0} + \mu_{1,2})^2 + (\mu_{0,3} + \mu_{2,1})^2 \\
\phi_5 =& (\mu_{3,0} - 3\mu_{1,2})(\mu_{3,0} + \mu_{1,2})\left[(\mu_{3,0} + \mu_{1,2})^2 - 3(\mu_{0,3} + \mu_{2,1})^2\right] \\
&+ (\mu_{0,3} - 3\mu_{2,1})(\mu_{0,3} + \mu_{2,1})\left[(\mu_{0,3} + \mu_{2,1})^2 - 3(\mu_{3,0} + \mu_{1,2})^2\right] \\
\phi_6 =& (\mu_{2,0} - \mu_{0,2})\left[(\mu_{3,0} + \mu_{1,2})^2 - (\mu_{0,3} + \mu_{2,1})^2\right] + 4\mu_{1,1}(\mu_{3,0} + \mu_{1,2})(\mu_{0,3} + \mu_{2,1}) \\
\phi_7 =& (3\mu_{2,1} - \mu_{0,3})(\mu_{3,0} + \mu_{1,2})\left[(\mu_{3,0} + \mu_{1,2})^2 - 3(\mu_{0,3} + \mu_{2,1})^2\right] \\
&+ (\mu_{3,0} - 3\mu_{2,1})(\mu_{0,3} + \mu_{2,1})\left[(\mu_{0,3} + \mu_{2,1})^2 - 3(\mu_{3,0} + \mu_{1,2})^2\right]
\end{aligned} \tag{33}$$

In addition to the seven invariant moments mentioned above, various methods exist for calculating shape-independent moments. In reference [61], Yang and Albregtsen introduced a method for efficiently calculating moments in binary images using Green's theorem. Since many effective invariants are derived from repeated experiments, Kapur et al. developed algorithms to systematically identify specific geometric invariants [62]. Gross and Latecki also devised a method that preserves the qualitative differential geometry of object edges during image digitization [62]. Furthermore, reference [63] discusses a framework of algebraic curves and invariants for representing complex objects in mixed scenes. This framework employs polynomial fitting to capture local geometric information and geometric invariants for object matching and recognition.

5.2.4. Shape Features Based on Inner Angles

In reference [64], a method for expressing shape features based on internal angles was proposed. Similarly to the Fourier descriptor, the object is first approximated as a polygon. The internal angles of polygons are crucial for shape representation and recognition and can be expressed as Intra-angle $= \{\alpha 1, \alpha 2, \ldots . \alpha n\}$.

Obviously, the shape description based on internal angles is independent of the shape's position, rotation, and size, making it well suited for image retrieval systems. Below is a series of definitions for shape features derived from internal angles:

Number of vertices: The more vertices a polygon has, the more complex its shape. It is reasonable to consider two shapes with different numbers of vertices as distinctly different shapes.

Internal Angle Mean: The average value of all internal angles of a polygon reflects its shape attributes to some extent. For example, the internal angle mean of a triangle is 60 degrees, which is notably different from the internal angle mean of a rectangle, which is 90 degrees.

Internal Angle Standard Deviation: The standard deviation of the internal angles of a polygon provides insight into the variability of the angles relative to the mean.

$$\delta = \sqrt{\sum_{i=1}^{n} (a_i - \bar{a})^2} \tag{34}$$

where $\bar{a}$ is the mean value of the interior angles, this standard deviation δ serves as a general descriptor of the polygon. The more regular the polygon, the smaller the value of δ. Therefore, it can be used to distinguish regular polygons from irregular ones.

Intra-angle Histogram: The angle range of 0° to 360° is first divided into k intervals, which serve as the k bins of the histogram. Then, the number of interior angles within each interval is counted. The resulting intra-angle histogram reflects the overall distribution of the interior angles.

We will take the calculation of the interior angle $\theta = \angle abc$ in Figure 4 as an example to introduce the calculation method of the interior angle, Figure 4 is a diagram of the interior angles of a plane graph. Let the center of the three points a, b, and c be p, then

$$o\vec{p} = \frac{o\vec{a} + o\vec{b} + o\vec{c}}{3} \tag{35}$$

where o is the origin. If p is inside the polygon, then θ is less than 180°; otherwise, θ is greater than 180°. When $\theta \leqslant 180°$,

$$\theta = \arccos\left(\frac{|ab|^2 + |bc|^2 - |ac|^2}{2|ab|bc\ |}\right) \tag{36}$$

除了上述七个不变矩之外，还存在多种用于计算形状无关矩的方法。在文献[61]中，Yang和Albregtsen提出了一种基于格林定理的高效算法，用于在二值图像中计算矩。由于许多有效的不变量是通过反复实验获得的，Kapur等人开发了能够系统性地识别特定几何不变量的算法[62]。此外，Gross和Latecki还提出了一种在图像数字化过程中保持目标边缘定性微分几何特性的方法[62]。更进一步地，文献[63]探讨了一个用于表示混合场景中复杂目标的代数曲线及其不变量框架；该框架采用多项式拟合来捕捉局部几何信息，并利用几何不变量进行目标匹配与识别。

5.2.4 基于内角的形状特征

在文献[64]中，提出了一种基于内角的形状特征表达方法。与傅里叶描述子类似，该方法首先将目标近似为一个多边形。多边形的内角对于形状的表示与识别至关重要，可表示为内角序列 = {α1, α2, ⋯, {vn}n。

显然，基于内角的形状描述与形状的位置、旋转和尺度无关，因此非常适用于图像检索系统。以下是若干由内角导出的形状特征的定义：

顶点数：多边形的顶点数越多，其形状就越复杂。因此，将具有不同顶点数的两个形状视为截然不同的形状是合理的。

内角均值：多边形所有内角的平均值在一定程度上反映了其形状特征。例如，三角形的内角均值为60度，这与矩形的内角均值90度有显著差异。

内角标准差：多边形各内角的标准差能够反映各内角相对于均值的离散程度。

$$\delta = \sqrt{\sum_{i=1}^{n}(a_i - \bar{a})^2} \tag{34}$$

其中，$\bar{a}$ 为内角的平均值，该标准差 δ 可作为多边形的一般性描述指标。多边形越规则，其值 δ 越小。因此，可用于区分规则多边形与不规则多边形。

内角直方图：首先将0°至360°的角域划分为k个区间，这些区间作为直方图的k个桶；随后统计每个区间内的内角数量。最终得到的内角直方图反映了内角的整体分布特征。

我们以图4中内角$\theta = \angle abc$的计算为例，介绍内角的计算方法。图4为平面图的内角示意图。设三点a、b和c的外接圆圆心为p，则

$$o\vec{p} = \frac{o\vec{a} + o\vec{b} + o\vec{c}}{3} \tag{35}$$

其中o为原点。若p位于多边形内部，则θ小于180°；否则，θ大于180°。当θ ⩽等于180°时，

$$\theta = \arccos\left(\frac{|ab|^2 + |bc|^2 - |ac|^2}{2|ab|bc\ |}\right) \tag{36}$$

When $\theta > 180°$,

$$\theta = 360 - \arccos\left(\frac{|ab|^2 + |bc|^2 - |ac|^2}{2|ab||bc|}\right) \quad (37)$$

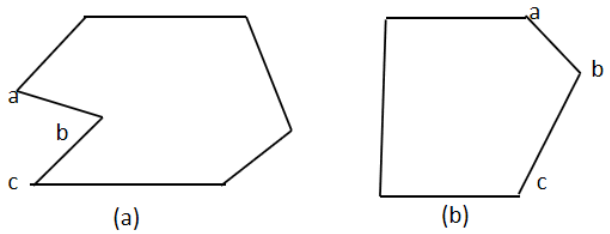


Figure 4. Interior angle histogram. (**a**) $\theta > 180°$, (**b**) $\theta \leqslant 180°$.

6. Experimental and Performance Comparison of Various Algorithms

6.1. Datasets

The commonly used datasets in the field of floor plan retrieval include SESYD, FPLAN-POLY, and HOME. These datasets are outlined as follows: Delalande et al. introduced the SESYD (Systems Evaluation Synthetic Documents) public dataset [65], a comprehensive document database synthesized using the 3gT (generation of graphical ground truth) system. This dataset is primarily employed in the GREC (Graphics Recognition) symbol recognition and discovery competition, comprising 10 classes of planar maps with 100 samples per class, totaling 28,065 symbols. The CVC-FP (Computer Vision Center Floor Plan) dataset, introduced in reference [66], encompasses four subsets: the Black dataset, Textured1 dataset, Textured2 dataset, and Parallel dataset. This dataset consists of 122 scanned documents and illustrates the relationship between architectural symbols and structural elements. Detailed labeling of these symbols facilitates more effective extraction of structural arrangements by floor plan analysis systems, thereby enhancing interpretive performance.

The FPLAN-POLY dataset [67] comprises 42 vector images featuring 38 symbolic models with 344 instances, assessing the presence of 8 structural conformations in each diagram. LV et al. [26] introduced the publicly available dataset ROBIN [68], which is specifically tailored for building floor plan retrieval tasks. ROBIN encompasses three primary layout categories, further subdivided into 10 subcategories based on global map layout shapes. Designed to cater to the needs of potential buyers, ROBIN showcases diverse room types and quantities.

FloorPlanCAD, a large vector dataset proposed by YUKI et al. [29], features 10,094 floor plan data entries derived from real-world sources, including homes, shopping malls, schools, and other structures. The dataset is divided into 6382 training plans and 3712 test plans. Similarly, the BRIDGE dataset, proposed by SHREYA et al. [69], consists of over 13,000 building floor plan images, accompanied by task-specific annotations such as symbols, region descriptions, and paragraph details.

Alain, P et al. introduced the comprehensive SFPI (Synthetic Floor Plan Images) dataset [21], which consists of 10,000 images encompassing 16 furniture categories and a total of 300,000 furniture symbols. To enhance the diversity of the dataset, data augmentation techniques, including random angle rotations, were applied. Detailed information about the dataset is provided in Table 2.

当 $\theta >$ 为 180° 时，

$$\theta = 360 - \arccos\left(\frac{|ab|^2 + |bc|^2 - |ac|^2}{2|ab||bc|}\right) \tag{37}$$

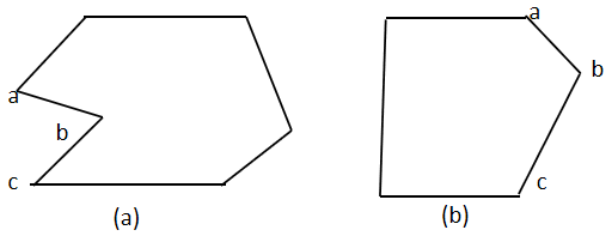


图4. 内角直方图。（a）$\theta > 180°$，（b）$\theta \leqslant 180°$。

6. 各种算法的实验与性能比较

6.1. Datasets

在平面图检索领域，常用的公开数据集包括SESYD、FPLAN-POLY和HOME。这些数据集的具体情况如下：Delalande等人提出了SESYD（Systems Evaluation Synthetic Documents）公开数据集[65]，这是一个基于3gT（generation of graphical ground truth）系统合成的综合性文档数据库。该数据集主要用于GREC（Graphics Recognition）符号识别与发现竞赛，包含10类平面图，每类100个样本，共计28,065个符号。此外，文献[66]中提出的CVC-FP（Computer Vision Center Floor Plan）数据集包含四个子集：Black数据集、Textured1数据集、Textured2数据集以及Parallel数据集。该数据集由122份扫描文档组成，展示了建筑符号与结构元素之间的对应关系。通过对这些符号进行详尽的标注，有助于平面图分析系统更有效地提取结构布局信息，从而提升其解释性能。

FPLAN-POLY数据集[67]包含42幅矢量图像，涵盖38种符号化模型及344个实例，用于评估每幅图中8种结构构型的存在情况。LV等人[26]提出了公开可用的数据集ROBIN[68]，该数据集专为建筑平面图检索任务而设计。ROBIN包含三大布局类别，并根据全局地图的布局形态进一步细分为10个小类。ROBIN旨在满足潜在购房者的需求，展示了多样化的房间类型与数量。

由YUKI等人[29]提出的大型矢量数据集FloorPlanCAD包含10,094个源自真实场景的平面图数据，涵盖住宅、购物中心、学校及其他建筑类型。该数据集被划分为6,382个训练用平面图和3,712个测试用平面图。与此类似，由SHREYA等人[69]提出的BRIDGE数据集包含超过13,000张建筑平面图图像，并附有针对特定任务的标注信息，如符号、区域描述及段落级细节。

Alain, P 等人提出了综合性的 SFPI（合成平面图图像）数据集[21]，该数据集包含 10,000 幅图像，涵盖 16 类家具及共计 300,000 个家具符号。为提升数据集的多样性，研究者采用了包括随机角度旋转在内的数据增强技术。有关该数据集的详细信息见表 2。

Table 2. Summary of datasets mentioned in this article.

Dataset Name	Number of Pictures	Usage	Public	Year
SESYD [65]	1000	Retrieval, Symbol localization	yes	2010
LIFULL HOME'S Dataset [70]	8300 ten thousand	Retrieval, Deep learning, Text mining	yes	2015
CVC-FP [71]	122	Semantic segmentation	yes	2015
FPLAN-POLY [72]	42	Symbolic positioning	yes	2010
ROBIN [68]	510	Retrieval, Symbol location	yes	2017
FloorPlanCAD [22]	10,094	Panoramic symbol positioning	yes	2021
BRIDGE [73]	13,000	Symbol recognition Scene map composition Retrieval	yes	2019
SFPI [74]	10,000	Building plan analysis Symbol positioning Building plan analysis	yes	2022

The datasets utilized in this study originate from various building types, including residential, commercial, and industrial sectors, which exhibit significant differences in both design and function. For instance, residential floor plans are typically simplistic, whereas commercial buildings may incorporate complex spatial layouts and functional zoning. Furthermore, these datasets encompass architectural designs from different historical periods, reflecting the evolution of architectural styles and technologies. For example, modern buildings tend to emphasize open spaces and sustainability, while traditional structures often prioritize symmetry and ornamentation. Datasets with a limited number of samples may not adequately represent all possible building types and design styles, thereby constraining the generalizability of the research findings. Additionally, the quality of annotations in the datasets, such as information regarding room types, sizes, and functions in the floor plans, significantly influences their representativeness. High-quality annotations enhance the practical applicability of the datasets, whereas low-quality annotations may adversely affect model training outcomes. In practical applications, it is essential to integrate datasets from diverse sources, types, and regions. Employing data augmentation techniques, such as rotation, scaling, and noise addition, can enhance the diversity of the datasets and improve the robustness of the model.

6.2. Evaluation

In the domain of building plan retrieval, evaluation metrics are critical for assessing the performance of retrieval systems. Commonly used evaluation metrics include accuracy, F1 score, mean average precision (mAP), and the Matthews correlation coefficient (MCC) [75]. The MCC exhibits superior performance in addressing category imbalance issues in datasets compared to metrics such as accuracy, precision, and recall. To comprehensively assess the effectiveness and utility of retrieval methods, additional evaluation techniques have been employed in some studies. In this context, a retrieval is deemed successful if one or more of the top five retrieved images exhibit structural similarity to the query image. Retrieval performance is quantified by calculating the proportion of correctly matched images within the retrieval results. For example, if three out of the first five retrieved images are structurally similar to the query image, the precision value is 0.6, indicating that 60% of the retrieved images are correct matches.

Different types of building floor plans, such as those for residential, office, and shopping mall purposes, exhibit varying degrees of structural complexity. For instance, residential floor plans are typically simpler, whereas shopping malls and office buildings tend to encompass a greater number of rooms, corridors, and intricate layouts. If the algorithm demonstrates strong performance on simpler floor plans but underperforms on more complex layouts, this indicates insufficient generalization capabilities. Complex floor plans

表2. 本文中提及的数据集汇总。

Dataset Name	Number of Pictures	Usage	Public	Year
SESYD [65]	1000	Retrieval, Symbol localization	yes	2010
LIFULL HOME'S Dataset [70]	8300 ten thousand	Retrieval, Deep learning, Text mining	yes	2015
CVC-FP [71]	122	Semantic segmentation	yes	2015
FPLAN-POLY [72]	42	Symbolic positioning	yes	2010
ROBIN [68]	510	Retrieval, Symbol location	yes	2017
FloorPlanCAD [22]	10,094	Panoramic symbol positioning	yes	2021
BRIDGE [73]	13,000	Symbol recognition Scene map composition Retrieval	yes	2019
SFPI [74]	10,000	Building plan analysis Symbol positioning Building plan analysis	yes	2022

本研究中所使用的数据集来源于多种建筑类型，包括住宅、商业和工业建筑，这些建筑在设计与功能上均存在显著差异。例如，住宅的平面布局通常较为简单，而商业建筑则可能采用复杂的空间布局与功能分区。此外，这些数据集还涵盖了不同历史时期的建筑设计，反映了建筑风格与技术的演变过程。例如，现代建筑往往强调开放空间与可持续性，而传统建筑则更注重对称性与装饰性。样本数量有限的数据集可能无法充分涵盖所有可能的建筑类型与设计风格，从而限制了研究结果的普适性。同时，数据集中标注的质量，如平面图中关于房间类型、面积及功能等信息的准确性，也会显著影响其代表性。高质量的标注能够提升数据集的实际应用价值，而低质量的标注则可能对模型训练效果产生不利影响。在实际应用中，有必要整合来自不同来源、不同类型及不同地区的数据集，并通过旋转、缩放、添加噪声等数据增强技术来提升数据集的多样性，进而提高模型的鲁棒性。

6.2. Evaluation

在建筑平面检索领域，评价指标对于评估检索系统的性能至关重要。常用的评价指标包括准确率、F1分数、平均精度均值（mAP）以及马修斯相关系数（MCC）[75]。与准确率、精确率和召回率等指标相比，MCC在处理数据集中的类别不平衡问题时表现出更优的性能。为全面评估检索方法的有效性和实用性，部分研究还采用了其他评价方法。在此背景下，若检索结果中排名前五的图像中有至少一张与查询图像在结构上相似，则认为该检索成功。检索性能通过计算检索结果中正确匹配图像所占的比例来量化。例如，若前五张检索到的图像中有三张与查询图像在结构上相似，则精确率为0.6，表明检索结果中60%的图像为正确匹配。

不同类型的建筑平面图，如住宅、办公和购物中心等，其结构复杂程度各不相同。例如，住宅平面图通常较为简单，而购物中心和办公楼则往往包含更多的房间、走廊以及更为复杂的布局。如果算法在较简单的平面图上表现良好，但在更复杂的布局上性能欠佳，则表明其泛化能力不足。复杂的建筑平面图

often present additional challenges such as increased occlusions, overlapping areas, and nonlinear structures, all of which can complicate feature extraction. Therefore, this paper recommends that experiments incorporate a diverse range of building floor plan types, with a thorough analysis of the performance for each category. Furthermore, for complex floor plans, the introduction of advanced feature extraction techniques, such as neural networks, or the implementation of data augmentation strategies, may enhance performance.

The resolution of building plans can vary significantly depending on the source. Low-resolution images may result in the loss of important details, while high-resolution images can increase computational complexity. This discrepancy may hinder the accurate recognition of key features, such as walls, doors, and windows, ultimately diminishing the performance of the algorithm. Therefore, this paper recommends evaluating the algorithm's performance across various resolutions during the experiments and analyzing the relationship between resolution and performance. Additionally, it may be beneficial to employ a multi-scale feature extraction method to effectively process both low-resolution and high-resolution images.

Secondly, with respect to noise, building plans may contain various forms of interference, such as blurred images, incorrect annotations, and extraneous text or symbols. Such noise can disrupt the feature extraction process of the algorithm, leading to false detections or missed detections. If the algorithm is sensitive to noise, its performance in practical applications may be significantly compromised. Therefore, this paper recommends incorporating a denoising step in the data preprocessing stage, utilizing image filtering techniques or deep learning denoising models.

6.3. Various Algorithms' Performance Comparison

While precision, recall, and mAP are established performance evaluation metrics in the existing literature, this study demonstrates system performance by evaluating feature extraction at each hierarchical level. The features extracted from the normalized Layer 1 exhibit strong performance, attaining an mAP value of 0.63. In contrast, the basic deep learning framework performs poorly when extracting features directly from the CNNs, yielding a mere average accuracy of 0.43. Furthermore, methods such as Scale-Invariant Feature Transform (SIFT), Histogram of Oriented Gradients (HOGs), run-length histograms, and OASIS yield suboptimal results in capturing the abstraction and sparsity inherent in sketches [76,77]. The comparative retrieval performance of various methods is presented in Table 3. Notably, the retrieval performance of the graph GCN method has significantly improved compared to previous techniques, attaining an mAP of 0.85. Furthermore, the latest YOLOv8-L achieves an mAP of 0.90 on the public COCO dataset.

In addition, we used several commonly used methods and compared the F1 value and mAp on the semantic features of different categories of plan views [22]. The experimental results are shown in Table 4.

From Table 4, it is evident that the F1 scores for the GCN-based and DccpLabv3 methods differ significantly, and their corresponding mAP values are relatively small. However, as the number of categories increases, both the F1 and mAP values tend to decrease. This suggests that the characteristics of the categories significantly influence retrieval performance.

此类数据往往还存在遮挡增多、区域重叠以及非线性结构等额外挑战，这些因素均会增加特征提取的难度。因此，本文建议在实验中纳入多种类型的建筑平面布局，并对各类别模型的性能进行深入分析。此外，针对复杂平面布局，引入诸如神经网络等先进的特征提取技术，或采用数据增强策略，有望进一步提升模型性能。

建筑平面图的分辨率因来源不同而存在显著差异。低分辨率图像可能导致重要细节的丢失，而高分辨率图像则会增加计算复杂度。这种分辨率差异可能妨碍对墙体、门、窗等关键特征的准确识别，从而最终降低算法的性能。因此，本文建议在实验过程中评估算法在不同分辨率下的性能，并分析分辨率与性能之间的关系。此外，采用多尺度特征提取方法以有效处理低分辨率和高分辨率图像也可能带来益处。

其次，就噪声而言，建筑图纸中可能存在多种类型的干扰，如图像模糊、标注错误以及多余的文本或符号。此类噪声会干扰算法的特征提取过程，导致误检或漏检。若算法对噪声较为敏感，则其在实际应用中的性能可能会受到显著影响。因此，本文建议在数据预处理阶段引入去噪步骤，采用图像滤波技术或深度学习去噪模型。

6.3. Various Algorithms′ Performance Comparison

尽管精确率、召回率和平均精度均值（mAP）在现有文献中已被确立为标准的性能评估指标，本研究通过评估各层次特征提取的性能来表征系统整体表现。从归一化后的第1层提取的特征表现出优异的性能，其mAP值达到0.63。相比之下，直接从卷积神经网络（CNN）提取特征的基本深度学习框架性能较差，平均精度仅为0.43。此外，尺度不变特征变换（SIFT）、方向梯度直方图（HOGs）、游程直方图以及OASIS等方法在捕捉素描固有的抽象性和稀疏性方面均未能取得理想效果[76,77]。各类方法的检索性能对比如表3所示。值得注意的是，图卷积网络（GCN）方法的检索性能较以往技术有了显著提升，mAP达到0.85。此外，最新的YOLOv8-L模型在公开的COCO数据集上实现了0.90的mAP。

此外，我们采用了若干常用方法，并比较了不同类别平面图语义特征的F1值和mAp[22]。实验结果如表4所示。

由表4可知，基于GCN的方法与DccpLabv3方法的F1分数存在显著差异，且两者的mAP值均相对较低。然而，随着类别数量的增加，F1分数和mAP值均呈现下降趋势，这表明类别的特征对检索性能具有显著影响。

Table 3. Building plan retrieval performance comparison.

Methodology	Dataset	Performance	Year
RLH + Chechik et al. [78]	ROBIN	mAp = 0.10	2009
RLH + Chechik et al. [78]	SESYD	mAp = 1.0	2009
BOW + Chechik et al. [78]	ROBIN	mAp = 0.19	2009
BOW + Chechik et al. [78]	SESYD	mAp = 1.0	2009
HOG + Chechik et al. [78]	ROBIN	mAp = 0.31	2009
DANIEL [68]	ROBIN	mAp = 0.56	2017
Sharma et al. [32]	ROBIN	mAp = 0.25	2016
CVPR [79]	ROBIN	mAp = 0.29	2016
MCS [79]	HOME	-	2018
CNNs(update) [43]	HOME	Accuracy = 0.49	2018
Sharma and Chattopadhyay [33]	ROBIN	mAp = 0.31	2018
Sharma and Chattopadhyay [33]	SESYD	mAp = 1.0	2018
FCNs [18]	HOME	mAp = 0.39	2018
REXplore [76]	ROBIN	mAp = 0.63	2018
Rasika et al. [52]	ROBIN	mAp = 0.74	2021
RISC-Net [77]	ROBIN	mAp = 0.79	2022
GCNs	ROBIN	mAp = 0.85	–
YOLOv8-L	COCO	mAp = 0.9	2024

Table 4. Comparison of semantic features of different categories of floor plans and F1 values and mAP of commonly used retrieval methods.

Class	Semantic Symbol Spotting		Instance Symbol Spotting		
	Weighted Fl		mAP		
	GCN-Based	DccpLabv3+	Faster R-CNN	FCOS	YOLOv3
single door	0.885	0.827	0.843	0.859	0.829
double door	0.796	0.831	0.779	0.771	0.743
sliding door	0.874	0.876	0.556	0.494	0.481
window	0.691	0.603	0.518	0.465	0.379
bay window	0.050	0.163	0.068	0.169	0.062
blind window	0.833	0.856	0.614	0.520	0.322
opening symbol	0.451	0.721	0.496	0.542	0.168
stairs	0.857	0.853	0.464	0.487	0.370
gas stove	0.789	0.847	0.503	0.715	0.601
refrigerator	0.705	0.730	0.767	0.774	0.723
washing machine	0.784	0.569	0.379	0.261	0.374
sofa	0.606	0.674	0.160	0.133	0.435
bed	0.893	0.908	0.713	0.738	0.664
chair	0.524	0.543	0.112	0.087	0.132
table	0.354	0.496	0.175	0.109	0.173

7. Conclusions

In the field of architectural floor plan feature retrieval, the integration of semantic, spatial, texture, and shape features has significantly improved the accuracy and efficiency of retrieval systems. Semantic features enable a deeper understanding of the functional aspects of floor plan elements, whereas spatial features clarify the relationships between

表3. 建筑方案检索性能对比。

Methodology	Dataset	Performance	Year
RLH + Chechik et al. [78]	ROBIN	mAp = 0.10	2009
RLH + Chechik et al. [78]	SESYD	mAp = 1.0	2009
BOW + Chechik et al. [78]	ROBIN	mAp = 0.19	2009
BOW + Chechik et al. [78]	SESYD	mAp = 1.0	2009
HOG + Chechik et al. [78]	ROBIN	mAp = 0.31	2009
DANIEL [68]	ROBIN	mAp = 0.56	2017
Sharma et al. [32]	ROBIN	mAp = 0.25	2016
CVPR [79]	ROBIN	mAp = 0.29	2016
MCS [79]	HOME	-	2018
CNNs(update) [43]	HOME	Accuracy = 0.49	2018
Sharma and Chattopadhyay [33]	ROBIN	mAp = 0.31	2018
Sharma and Chattopadhyay [33]	SESYD	mAp = 1.0	2018
FCNs [18]	HOME	mAp = 0.39	2018
REXplore [76]	ROBIN	mAp = 0.63	2018
Rasika et al. [52]	ROBIN	mAp = 0.74	2021
RISC-Net [77]	ROBIN	mAp = 0.79	2022
GCNs	ROBIN	mAp = 0.85	–
YOLOv8-L	COCO	mAp = 0.9	2024

表4. 不同类别平面图的语义特征比较及常用检索方法的F1值与mAP。

Class	Semantic Symbol Spotting		Instance Symbol Spotting		
	Weighted Fl		mAP		
	GCN-Based	DccpLabv3+	Faster R-CNN	FCOS	YOLOv3
single door	0.885	0.827	0.843	0.859	0.829
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window	0.691	0.603	0.518	0.465	0.379
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opening symbol	0.451	0.721	0.496	0.542	0.168
stairs	0.857	0.853	0.464	0.487	0.370
gas stove	0.789	0.847	0.503	0.715	0.601
refrigerator	0.705	0.730	0.767	0.774	0.723
washing machine	0.784	0.569	0.379	0.261	0.374
sofa	0.606	0.674	0.160	0.133	0.435
bed	0.893	0.908	0.713	0.738	0.664
chair	0.524	0.543	0.112	0.087	0.132
table	0.354	0.496	0.175	0.109	0.173

7. 结论

在建筑平面图特征检索领域，语义、空间、纹理及形状特征的融合显著提升了检索系统的准确性和效率。语义特征有助于深入理解平面图各要素的功能属性，而空间特征则阐明了各要素之间的关系。

different building layouts. Texture features improve the visual consistency of floor plans, while shape features aid in the accurate identification of various architectural styles. This paper thoroughly examines these feature retrieval methods and introduces advanced fusion techniques to support future research, with the goal of achieving greater automation and intelligence in retrieval systems. Ultimately, the findings are expected to drive advancements in architectural floor plan data processing technology and promote its practical application.

We believe that the following are potential directions for floor plan retrieval in the future:

1. Diverse graphic retrieval and cross-application, encompassing images, text, audio, video, and other modalities, are becoming increasingly relevant.
   Future building design tools are expected to prioritize user experience by integrating various data sources for federated search, thereby providing more comprehensive and accurate results. By incorporating personalized needs and collaborative design into the process, user satisfaction and efficiency can be significantly improved. Future research should explore multimodal data representation, processing, and retrieval, combining floor plans with other modalities such as text, 3D models, and satellite images to achieve a holistic understanding of building design. Additionally, integrating data from building design software (e.g., Revit, AutoCAD) with parametric design tools (e.g., Grasshopper) will create a unified data platform that enables real-time updates and sharing of design information. Constructing a homogeneous graph network model of building structures, where nodes represent components (such as beams, columns, and walls) and edges illustrate the relationships between them, will enhance structural analysis. In the domains of smart city planning, architectural design, and virtual reality (VR), BIM and VR-enhanced visualization and interaction capabilities enable improved design coordination, reducing problem-solving time by 15% and enhancing stakeholder engagement and satisfaction by 20%. Rule-based model checking and generative design methods are employed to ensure compliance with urban development and sustainability standards, delivering environmentally sustainable and cost-effective design solutions. Multidimensional feature retrieval of floor plans will become increasingly important to achieve accurate design, simulation, and optimization. Creating immersive models will enable stakeholders to intuitively understand spatial and functional layouts during the design phase. This study showed that the use of BIM and VR significantly enhanced communication, collaboration, and decision-making among project stakeholders. A detailed taxonomy of AR interactions was combined with a frequency analysis, where navigation interactions were the most common, accounting for 62% of all interactions, followed by preparation interactions at 22%, annotation interactions at 10%, and recording interactions at 6%. Transitions between different design artifacts are critical to solving design problems, with 47% of transitions occurring in the transition from 2D digital information (PDF) to other artifacts, 30% in 3D digital information, and 23% in physical drawings [80]. These findings indicate that effective management of digital transformation is critical to optimizing design workflows.
2. Efficient feature extraction, indexing, and personalized adaptive retrieval are critical components.
   As data volumes grow, the efficient indexing and retrieval of multidimensional features have become indispensable. Future research could prioritize the development of efficient and lightweight deep learning models capable of real-time multidimensional feature extraction on mobile and embedded systems. By leveraging user interaction data, future floor plan retrieval systems can adaptively adjust feature weights and optimization strategies to deliver personalized retrieval outcomes. Even with minimal

不同的建筑布局。纹理特征有助于提升平面图的视觉一致性，而形状特征则有助于准确识别各类建筑风格。本文深入探讨了这些特征检索方法，并引入了先进的融合技术，以支持后续研究，旨在实现检索系统的更高自动化与智能化。最终，研究成果有望推动建筑平面图数据处理技术的进步，并促进其实际应用。

我们认为，未来楼层平面图检索的潜在研究方向如下：

1. 多模态的图文检索与跨应用集成，涵盖图像、文本、音频、视频等多种模态，正变得日益重要。

未来建筑设计工具有望通过整合多种数据源实现联邦搜索，从而优先考虑用户体验，提供更为全面且准确的检索结果。通过在设计过程中融入个性化需求与协同设计理念，用户的满意度与设计效率将得到显著提升。未来的研究应深入探索多模态数据的表征、处理与检索方法，将建筑平面图与其他模态（如文本、三维模型及卫星影像）相结合，以实现对建筑设计的全局性理解。此外，将建筑设计软件（如Revit、AutoCAD）与参数化设计工具（如Grasshopper）的数据相融合，可构建统一的数据平台，支持设计信息的实时更新与共享。通过构建建筑结构的同质图网络模型，其中节点代表构件（如梁、柱、墙等），边则刻画各构件之间的关联关系，有助于强化结构分析能力。在智慧城市规划、建筑设计以及虚拟现实（VR）等领域，基于BIM与VR的可视化与交互功能能够提升设计协调水平，使问题解决时间缩短15%，并使利益相关方的参与度与满意度提升20%。采用基于规则的模型检查与生成式设计方法，可确保设计方案符合城市开发与可持续性标准，从而提供环境友好且经济高效的设计方案。为实现精准的设计、仿真与优化，建筑平面图的多维特征检索将愈发重要。构建沉浸式模型则能使利益相关方在设计阶段直观地理解空间布局与功能分区。本研究证实，BIM与VR的应用显著提升了项目各参与方之间的沟通、协作与决策效率。通过对AR交互行为进行详细分类并开展频次分析发现，导航类交互最为常见，占所有交互的62%；其次是准备类交互，占比22%；注释类交互占10%；记录类交互占6%。不同设计成果之间的转换对于解决设计问题至关重要，其中47%的转换发生在从二维数字信息（PDF）向其他成果的过渡中，30%发生在三维数字信息之间，23%发生在实体图纸之间[80]。这些研究结果表明，有效管理数字化转型是优化设计工作流程的关键。

2. 高效的特征提取、索引构建以及个性化自适应检索是关键组成部分。

随着数据规模的不断增长，多维特征的高效索引与检索已变得不可或缺。未来的研究可优先致力于开发能够在移动设备和嵌入式系统上实现多维特征实时提取的高效轻量化深度学习模型。通过利用用户交互数据，未来的楼层平面图检索系统能够自适应地调整特征权重及优化策略，从而提供个性化的检索结果。即便在资源受限的情况下

user feedback, the retrieval model can be iteratively refined, significantly improving system performance.

3. Interpretability of multidimensional data and data privacy security are critical considerations.
   As deep learning models grow in complexity, ensuring interpretability has become increasingly important. Future research could focus on developing interpretable models that elucidate the relationship between multidimensional features and retrieval results, thereby enabling users to comprehend the system's decision-making process. Moreover, with growing concerns over data privacy, it is imperative to ensure efficient floor plan retrieval while safeguarding user privacy. Technologies such as differential privacy and federated learning must be incorporated into retrieval systems. Furthermore, protecting sensitive architectural and design data from unauthorized access will become a critical research focus, especially in the context of military and government building designs.

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通过用户反馈，检索模型可以进行迭代优化，从而显著提升系统性能。

3. 多维数据的可解释性以及数据隐私安全是至关重要的考量因素。

随着深度学习模型复杂度的不断提升，确保其可解释性已变得愈发重要。未来的研究可着重于构建能够揭示多维特征与检索结果之间关系的可解释模型，从而帮助用户理解系统的决策过程。此外，鉴于数据隐私问题日益受到关注，在保障用户隐私的前提下实现高效的建筑平面图检索尤为必要。差分隐私与联邦学习等技术应被纳入检索系统的设计之中。进一步地，在军事及政府建筑设计等领域，如何保护敏感的建筑与设计数据免遭未经授权的访问也将成为一项重要的研究重点。

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机构审查委员会声明：不适用。

知情同意声明：不适用。

数据可用性声明：有关详情，请参阅网址 https://github.com/BYFCJX/download-floorplan-image-datasets 中的文件（访问日期：2025年3月16日）。

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